

COMPLETE

BIOLOGY

Upper Secondary / Terminale Coursebook

Biology • Cells • Human Biology • Genetics • Ecology

30 Complete Lessons • Practice Activities
Revision Tests • Answer Key • Reference Tables

A structured 80-page learning book for classroom teaching, independent study and exam preparation.

BIOLOGY SCIENCE SERIES

Course focus	Core scientific knowledge, interpretation, laboratory reasoning and exam skills.
Level	Upper Secondary / Terminale / Scientific stream.
Language	English.

GETTING STARTED

How to Use This Book

A practical roadmap for learning Biology step by step.

What You Will Learn

- Core Biology from cells and biological molecules to genetics, evolution and ecosystems.
- Accurate scientific vocabulary for school, examinations and practical work.
- Cause-and-effect explanations linking biological structure, process and function.
- Data interpretation through tables, model statements, guided practice and revision tests.

Recommended Study Method

1. Read the lesson objectives before studying the concepts.
2. Study the concept table and explain each example in your own words.
3. Complete guided practice without looking at the answers.
4. Correct every error by writing the scientific principle behind the correction.
5. Review difficult lessons after 24 hours and again after one week.

Biology Answering Checklist

- Command word: Did I state, describe, explain, compare or evaluate as requested?
- Vocabulary: Did I use the correct biological terms?
- Cause and effect: Did I explain how or why, not only what?
- Data: Did I quote values and units where needed?
- Diagram: Are labels clear, horizontal and scientifically correct?
- Conclusion: Is it supported by evidence?

Book Organization

- Lessons 1-10: Foundations, cells, transport, molecules, enzymes, plant and human nutrition.
- Lessons 11-20: Respiration, gas exchange, transport, excretion, coordination and reproduction.
- Lessons 21-30: Growth, DNA, cell division, genetics, evolution, ecology and conservation.
- Pages 65-72: Eight mixed revision tests.
- Pages 73-78: Answer keys, skills and definitions; pages 79-80: reference guide.

Suggested 6-Week Study Schedule

- Week 1: Lessons 1-5 - life processes, cells and biological molecules.
- Week 2: Lessons 6-10 - enzymes, plants and digestion.
- Week 3: Lessons 11-15 - respiration, circulation and plant transport.
- Week 4: Lessons 16-20 - excretion, homeostasis, coordination and reproduction.
- Week 5: Lessons 21-25 - growth, DNA, cell division and inheritance.
- Week 6: Lessons 26-30 + revision tests and error review.

Lessons 1-15

Part I Learning Map

- ## Part I Milestones

- I can identify structures and relate them to functions.
- I can explain transport and energy-releasing processes.
- I can use scientific vocabulary accurately.
- I can answer structured questions using cause-and-effect reasoning.

[illegible]

REFERENCE

Table of Contents

Lessons 16-30, Revision and Reference

Lesson	Topic	Pages
16	Excretion	35-36
17	Homeostasis	37-38
18	Nervous System	39-40
19	Hormones and Endocrine System	41-42
20	Human Reproduction	43-44
21	Plant Reproduction	45-46
22	Growth and Development	47-48
23	DNA, Genes and Chromosomes	49-50
24	Cell Division: Mitosis and Meiosis	51-52
25	Genetics and Inheritance	53-54
26	Variation and Natural Selection	55-56
27	Evolution	57-58
28	Ecology and Ecosystems	59-60
29	Food Chains, Food Webs and Nutrient Cycles	61-62
30	Human Impact, Conservation and Biology Review	63-64
Revision	Review Tests 1-8	65-72
Answers	Answer Key and Skills	73-78
Reference	Biology Reference Tables	79
Reference	Biology Quick Guide	80

Part II Learning Map

- Lessons 16-20: Excretion, homeostasis, coordination and reproduction.
- Lessons 21-25: Growth, DNA, mitosis, meiosis and inheritance.
- Lessons 26-30: Variation, evolution, ecology, cycles and conservation.

Part II Milestones

- I can explain regulation and coordination in organisms.
- I can use genetic terminology and basic inheritance models.
- I can explain evolution and ecological interactions.
- I can evaluate human impacts and conservation responses.

Part II Study Notes

LESSON 01 - CORE BIOLOGY

Introduction to Biology and Characteristics of Living Things

Concepts, functions, processes and model examples

Learning Objectives

- Define Biology and explain its scope.
- Identify the main characteristics of living organisms.
- Distinguish living organisms from non-living objects.

Concept	Explanation	Example
Biology	The scientific study of living organisms and life processes.	Study of plants, animals and microorganisms.
Movement	A change in position of an organism or part of it.	A bird flies; a shoot bends toward light.
Respiration	Chemical reactions in cells that release energy from nutrients.	Glucose is broken down to release energy.
Sensitivity	Detection of and response to changes in the environment.	A hand is withdrawn from a hot surface.
Growth	A permanent increase in size and dry mass.	A seedling develops into a mature plant.

Key Points

- A useful memory aid is MRS GREN: Movement, Respiration, Sensitivity, Growth, Reproduction, Excretion, Nutrition.
- Living organisms are cellular and carry out coordinated life processes.
- Biology supports medicine, agriculture, conservation and biotechnology.

Study Tip

Read each explanation, connect the structure to its function, and then create one new biological example using the same principle.

Extra Model Statements

1. A plant growing toward light shows sensitivity.
2. A child increasing in height shows growth.
3. Removing urea from the body is excretion.
4. Eating provides nutrients needed for energy and repair.

Scientific Analysis Routine

Step 1: Identify the biological structure, process or variable.

Step 2: State the relevant scientific principle.

Step 3: Link structure to function or cause to effect.

Step 4: Support the explanation with a clear example or evidence.

Your Example

Write one accurate biological example and underline the key scientific term.

LESSON 01 - PRACTICE

Introduction to Biology and Characteristics of Living Things

Common misconceptions, guided practice and mastery check

Common error	Better scientific idea	Why?
Plants do not move.	Plants show movements such as tropisms and opening or closing structures.	Movement in Biology is not limited to walking.
Respiration means breathing.	Respiration is a cellular chemical process; breathing is ventilation.	The two processes are related but not identical.

Guided Practice

1. What is Biology?
2. Name the seven characteristics of living things.
3. Give one example of sensitivity.
4. State one difference between excretion and egestion.
5. Which branch studies plants?
6. Which branch studies heredity?
7. Give two examples of metabolic waste.
8. Why is nutrition important?

Mastery Challenge

- Write two original explanations demonstrating the main ideas from Introduction to Biology and Characteristics of Living Things.
- Use correct biological vocabulary and connect evidence to the conclusion.
- Check units, labels and scientific spellings before comparing with the answers.

Quick Answers

1. The scientific study of life and living organisms.
2. Movement, respiration, sensitivity, growth, reproduction, excretion and nutrition.
3. For example, withdrawing a hand from heat.
4. Excretion removes metabolic waste; egestion removes undigested food.
5. Botany.
6. Genetics.
7. For example, carbon dioxide and urea.
8. It provides energy and materials for growth and repair.

Independent Practice

- A. Write three accurate statements using the main concepts from this lesson.
- B. Draw and label one relevant biological diagram or process flow where appropriate.
- C. Write one deliberately incorrect statement, then correct it and explain the scientific reason.
- D. Apply the lesson to a short real-life or laboratory context of 40-60 words.

Exam Strategy

- Read the command word carefully: state, describe, explain, compare, calculate or evaluate.
- Use precise scientific terms and show the link between cause and effect.
- For data questions, quote relevant evidence and include units.

Notes & Corrections

Record one mistake, the correction and the biological principle.

LESSON 02 - CORE BIOLOGY

Cell Structure and Cell Organization

Concepts, functions, processes and model examples

Learning Objectives

- Explain the cell as the basic unit of life.
- Identify major cell structures and their functions.
- Describe levels of organization from cell to organism.

Concept	Explanation	Example
Cell membrane	A selectively permeable boundary controlling movement of substances.	Controls entry of nutrients and exit of wastes.
Cytoplasm	Jelly-like region where many metabolic reactions occur.	Contains enzymes for cell reactions.
Nucleus	Contains genetic material and controls many cell activities.	Houses chromosomes in eukaryotic cells.
Mitochondrion	Site of most aerobic respiration.	Releases energy for active processes.
Ribosome	Site of protein synthesis.	Builds proteins from amino acids.

Key Points

- Cells are the basic structural and functional units of living organisms.
- In multicellular organisms: cells form tissues, tissues form organs, organs form systems.
- Structure is linked to function in specialized cells.

Study Tip

Read each explanation, connect the structure to its function, and then create one new biological example using the same principle.

Extra Model Statements

1. Muscle cells contain many mitochondria because contraction needs energy.
2. Red blood cells are specialized to transport oxygen.
3. An organ contains several tissues working together.
4. The digestive system contains several organs.

Scientific Analysis Routine

Step 1: Identify the biological structure, process or variable.

Step 2: State the relevant scientific principle.

Step 3: Link structure to function or cause to effect.

Step 4: Support the explanation with a clear example or evidence.

Your Example

Write one accurate biological example and underline the key scientific term.

LESSON 02 - PRACTICE

Cell Structure and Cell Organization

Common misconceptions, guided practice and mastery check

Common error	Better scientific idea	Why?
All cells have a nucleus.	Mature mammalian red blood cells do not have a nucleus.	Cell specialization can involve loss of some structures.
An organ is made of one type of cell.	Organs contain several tissues, and tissues contain cells.	Biological organization is hierarchical.

Guided Practice

1. Define a cell.
2. Name two structures found in most animal cells.
3. What is the function of mitochondria?
4. What is the function of ribosomes?
5. Arrange: organ, tissue, cell, organ system.
6. Give one example of a specialized cell.
7. What is a tissue?
8. Why do muscle cells contain many mitochondria?

Mastery Challenge

- Write two original explanations demonstrating the main ideas from Cell Structure and Cell Organization.
- Use correct biological vocabulary and connect evidence to the conclusion.
- Check units, labels and scientific spellings before comparing with the answers.

Quick Answers

1. The basic structural and functional unit of life.
2. For example, cell membrane and cytoplasm.
3. They are the main site of aerobic respiration.
4. They synthesize proteins.
5. Cell -> tissue -> organ -> organ system.
6. For example, sperm cell, neuron or red blood cell.
7. A group of similar cells working together.
8. They need large amounts of energy for contraction.

Independent Practice

- A. Write three accurate statements using the main concepts from this lesson.
- B. Draw and label one relevant biological diagram or process flow where appropriate.
- C. Write one deliberately incorrect statement, then correct it and explain the scientific reason.
- D. Apply the lesson to a short real-life or laboratory context of 40-60 words.

Exam Strategy

- Read the command word carefully: state, describe, explain, compare, calculate or evaluate.
- Use precise scientific terms and show the link between cause and effect.
- For data questions, quote relevant evidence and include units.

Notes & Corrections

Record one mistake, the correction and the biological principle.

LESSON 03 - CORE BIOLOGY

Plant and Animal Cells

Concepts, functions, processes and model examples

Learning Objectives

- Compare typical plant and animal cells.
- Identify plant-specific structures.
- Relate cell structures to their functions.

Concept	Explanation	Example
Cell wall	A cellulose layer that supports and protects plant cells.	Helps maintain cell shape.
Chloroplast	Organelle containing chlorophyll for photosynthesis.	Absorbs light energy.
Vacuole	Large fluid-filled sac in many plant cells.	Contains cell sap and helps maintain turgor.
Animal cell	Usually lacks cell wall and chloroplasts.	Cheek cell.
Plant cell	Usually has cell wall, chloroplasts and a large vacuole.	Leaf palisade cell.

Key Points

- Both plant and animal cells have a cell membrane, cytoplasm, ribosomes, mitochondria and genetic material.
- Only photosynthetic plant cells contain chloroplasts.
- The cell wall provides mechanical support but does not control exchange like the membrane does.

Study Tip

Read each explanation, connect the structure to its function, and then create one new biological example using the same principle.

Extra Model Statements

1. A palisade cell contains many chloroplasts.
2. A root hair cell usually lacks chloroplasts because it is underground.
3. Animal cells can change shape more easily because they lack a rigid cellulose wall.
4. Plant vacuoles help cells remain turgid.

Scientific Analysis Routine

Step 1: Identify the biological structure, process or variable.

Step 2: State the relevant scientific principle.

Step 3: Link structure to function or cause to effect.

Step 4: Support the explanation with a clear example or evidence.

Your Example

Write one accurate biological example and underline the key scientific term.

LESSON 03 - PRACTICE

Plant and Animal Cells

Common misconceptions, guided practice and mastery check

Common error	Better scientific idea	Why?
All plant cells contain chloroplasts.	Only photosynthetic plant cells normally contain chloroplasts.	Roots are an important exception.
The cell wall controls what enters the cell.	The cell membrane regulates movement of substances.	The wall is freely permeable to many substances.

Guided Practice

1. Name three structures present in plant cells but absent from typical animal cells.
2. What is the cell wall made of?
3. What is the function of chloroplasts?
4. Why do root hair cells usually lack chloroplasts?
5. Name two structures shared by plant and animal cells.
6. What is cell sap?
7. What is turgor?
8. Give one difference between a wall and a membrane.

Mastery Challenge

- Write two original explanations demonstrating the main ideas from Plant and Animal Cells.
- Use correct biological vocabulary and connect evidence to the conclusion.
- Check units, labels and scientific spellings before comparing with the answers.

Quick Answers

1. Cell wall, chloroplasts and a large permanent vacuole.
2. Cellulose.
3. They absorb light energy for photosynthesis.
4. They are usually underground and receive little light.
5. For example, membrane and mitochondria.
6. The solution inside the vacuole.
7. Internal pressure that supports a plant cell.
8. The membrane is selectively permeable; the wall mainly supports the cell.

Independent Practice

- A. Write three accurate statements using the main concepts from this lesson.
- B. Draw and label one relevant biological diagram or process flow where appropriate.
- C. Write one deliberately incorrect statement, then correct it and explain the scientific reason.
- D. Apply the lesson to a short real-life or laboratory context of 40-60 words.

Exam Strategy

- Read the command word carefully: state, describe, explain, compare, calculate or evaluate.
- Use precise scientific terms and show the link between cause and effect.
- For data questions, quote relevant evidence and include units.

Notes & Corrections

Record one mistake, the correction and the biological principle.

LESSON 04 - CORE BIOLOGY

Cell Membrane and Transport

Concepts, functions, processes and model examples

Learning Objectives

- Explain diffusion, osmosis and active transport.
- Predict movement of water and solutes across membranes.
- Relate transport processes to cell survival.

Concept	Explanation	Example
Diffusion	Net movement of particles from higher to lower concentration.	Oxygen diffuses from alveoli into blood.
Osmosis	Net movement of water through a partially permeable membrane from higher water potential to lower water potential.	Water enters root hair cells.
Active transport	Movement against a concentration gradient using cellular energy.	Mineral ions enter roots from dilute soil solution.
Concentration gradient	Difference in concentration between two regions.	A steeper gradient increases diffusion rate.
Partially permeable membrane	Allows some substances to cross more easily than others.	Cell membrane.

Key Points

- Diffusion and osmosis are passive processes; active transport requires energy.
- Temperature, surface area and concentration gradient affect diffusion rate.
- Plant cells become turgid in dilute solutions and plasmolysed in very concentrated solutions.

Study Tip

Read each explanation, connect the structure to its function, and then create one new biological example using the same principle.

Extra Model Statements

1. Oxygen diffuses into respiring cells.
2. Water moves into plant roots by osmosis.
3. Mineral ions can be absorbed by active transport.
4. A larger surface area increases the rate of exchange.

Scientific Analysis Routine

Step 1: Identify the biological structure, process or variable.

Step 2: State the relevant scientific principle.

Step 3: Link structure to function or cause to effect.

Step 4: Support the explanation with a clear example or evidence.

Your Example

Write one accurate biological example and underline the key scientific term.

LESSON 04 - PRACTICE

Cell Membrane and Transport

Common misconceptions, guided practice and mastery check

Common error	Better scientific idea	Why?
Osmosis is movement of any molecule.	Osmosis refers specifically to water movement through a partially permeable membrane.	Solute diffusion is not osmosis.
Active transport moves substances only from high to low concentration.	Active transport can move substances from low to high concentration.	Energy is used to work against the gradient.

Guided Practice

1. Define diffusion.
2. Define osmosis.
3. Which transport process needs energy?
4. Give one example of active transport in plants.
5. What happens to a plant cell in a dilute solution?
6. Name two factors that increase diffusion rate.
7. Why are membranes called selectively permeable?
8. What is a concentration gradient?

Mastery Challenge

- Write two original explanations demonstrating the main ideas from Cell Membrane and Transport.
- Use correct biological vocabulary and connect evidence to the conclusion.
- Check units, labels and scientific spellings before comparing with the answers.

Quick Answers

1. Net movement from high to low concentration.
2. Net movement of water through a partially permeable membrane from higher to lower water potential.
3. Active transport.
4. Uptake of mineral ions by root hair cells.
5. It becomes turgid.
6. For example, higher temperature and steeper gradient.
7. They allow some substances to cross more easily than others.
8. A difference in concentration between two regions.

Independent Practice

- A. Write three accurate statements using the main concepts from this lesson.
- B. Draw and label one relevant biological diagram or process flow where appropriate.
- C. Write one deliberately incorrect statement, then correct it and explain the scientific reason.
- D. Apply the lesson to a short real-life or laboratory context of 40-60 words.

Exam Strategy

- Read the command word carefully: state, describe, explain, compare, calculate or evaluate.
- Use precise scientific terms and show the link between cause and effect.
- For data questions, quote relevant evidence and include units.

Notes & Corrections

Record one mistake, the correction and the biological principle.

LESSON 05 - CORE BIOLOGY

Biological Molecules

Concepts, functions, processes and model examples

Learning Objectives

- Describe carbohydrates, proteins and lipids.
- State major functions of each nutrient group.
- Interpret common food tests.

Concept	Explanation	Example
Carbohydrates	Organic compounds including sugars and starches; major energy sources.	Glucose, starch, glycogen.
Proteins	Polymers of amino acids used for growth, repair and enzymes.	Muscle proteins and enzymes.
Lipids	Fats and oils used for energy storage, insulation and membranes.	Triglycerides and phospholipids.
Reducing sugar test	Benedict's reagent is heated with sample.	Positive result: green to brick-red precipitate.
Protein test	Biuret reagent is added to sample.	Positive result: purple/lilac.

Key Points

- Carbon, hydrogen and oxygen are common in carbohydrates and lipids; proteins also contain nitrogen.
- Excess nutrients can be stored or converted depending on the organism.
- Food tests indicate the presence of nutrient groups but not their exact quantity.

Study Tip

Read each explanation, connect the structure to its function, and then create one new biological example using the same principle.

Extra Model Statements

1. Starch stores carbohydrate in plants.
2. Glycogen stores carbohydrate in animals.
3. Proteins are needed for growth and tissue repair.
4. Lipids provide long-term energy storage.

Scientific Analysis Routine

Step 1: Identify the biological structure, process or variable.

Step 2: State the relevant scientific principle.

Step 3: Link structure to function or cause to effect.

Step 4: Support the explanation with a clear example or evidence.

Your Example

Write one accurate biological example and underline the key scientific term.

LESSON 05 - PRACTICE

Biological Molecules

Common misconceptions, guided practice and mastery check

Common error	Better scientific idea	Why?
Proteins are mainly used as the first energy source.	Carbohydrates are usually the immediate energy source; proteins have many structural and functional roles.	Proteins can be respired but are not normally the first choice.
The iodine test detects glucose.	Iodine detects starch.	Benedict's solution is used for reducing sugars.

Guided Practice

1. Name the building blocks of proteins.
2. Give two functions of lipids.
3. Which reagent tests for starch?
4. What color shows a positive iodine test?
5. Which test identifies reducing sugars?
6. Name one storage carbohydrate in animals.
7. Give one function of proteins.
8. Why are food tests described as qualitative?

Mastery Challenge

- Write two original explanations demonstrating the main ideas from Biological Molecules.
- Use correct biological vocabulary and connect evidence to the conclusion.
- Check units, labels and scientific spellings before comparing with the answers.

Quick Answers

1. Amino acids.
2. For example, energy storage and insulation.
3. Iodine solution.
4. Blue-black.
5. Benedict's test with heating.
6. Glycogen.
7. Growth and repair, or acting as enzymes.
8. They mainly show presence or absence rather than an exact amount.

Independent Practice

- A. Write three accurate statements using the main concepts from this lesson.
- B. Draw and label one relevant biological diagram or process flow where appropriate.
- C. Write one deliberately incorrect statement, then correct it and explain the scientific reason.
- D. Apply the lesson to a short real-life or laboratory context of 40-60 words.

Exam Strategy

- Read the command word carefully: state, describe, explain, compare, calculate or evaluate.
- Use precise scientific terms and show the link between cause and effect.
- For data questions, quote relevant evidence and include units.

Notes & Corrections

Record one mistake, the correction and the biological principle.

LESSON 06 - CORE BIOLOGY

Enzymes

Concepts, functions, processes and model examples

Learning Objectives

- Define enzymes as biological catalysts.
- Explain effects of temperature and pH on enzyme activity.
- Interpret enzyme-rate graphs.

Concept	Explanation	Example
Enzyme	A protein catalyst that speeds up a biological reaction without being used up.	Amylase breaks down starch.
Active site	Region of an enzyme where a substrate binds.	Shape is complementary to substrate.
Substrate	The reactant acted on by an enzyme.	Starch is a substrate for amylase.
Optimum	Condition at which enzyme activity is highest.	Human enzymes often work well near body temperature.
Denaturation	Permanent change in enzyme shape causing loss of function.	High temperature may alter the active site.

Key Points

- Enzymes are specific because the active site fits particular substrates.
- Rate rises with temperature up to an optimum, then falls sharply if enzymes denature.
- Extreme pH can alter bonds that maintain enzyme shape.

Study Tip

Read each explanation, connect the structure to its function, and then create one new biological example using the same principle.

Extra Model Statements

1. Amylase converts starch to maltose.
2. Protease digests proteins into smaller peptides or amino acids.
3. Lipase digests lipids into fatty acids and glycerol.
4. Catalase breaks down hydrogen peroxide.

Scientific Analysis Routine

Step 1: Identify the biological structure, process or variable.

Step 2: State the relevant scientific principle.

Step 3: Link structure to function or cause to effect.

Step 4: Support the explanation with a clear example or evidence.

Your Example

Write one accurate biological example and underline the key scientific term.

LESSON 06 - PRACTICE

Enzymes

Common misconceptions, guided practice and mastery check

Common error	Better scientific idea	Why?
Enzymes are used up during reactions.	Enzymes are catalysts and can be reused.	They are not consumed like reactants.
Higher temperature always increases enzyme activity.	Activity increases only up to an optimum; excessive heat can denature enzymes.	Shape determines function.

Guided Practice

1. Define an enzyme.
2. What is a substrate?
3. What happens at the active site?
4. Why can high temperature stop enzyme activity?
5. What is an optimum temperature?
6. Name the enzyme that digests starch.
7. Name the enzyme that digests lipids.
8. Why are enzymes described as specific?

Mastery Challenge

- Write two original explanations demonstrating the main ideas from Enzymes.
- Use correct biological vocabulary and connect evidence to the conclusion.
- Check units, labels and scientific spellings before comparing with the answers.

Quick Answers

1. A biological catalyst, usually a protein.
2. The substance acted on by an enzyme.
3. The substrate binds and is converted into product(s).
4. The enzyme may denature and its active site changes shape.
5. The temperature at which the rate is highest.
6. Amylase.
7. Lipase.
8. Each enzyme has an active site suited to particular substrate(s).

Independent Practice

- A. Write three accurate statements using the main concepts from this lesson.
- B. Draw and label one relevant biological diagram or process flow where appropriate.
- C. Write one deliberately incorrect statement, then correct it and explain the scientific reason.
- D. Apply the lesson to a short real-life or laboratory context of 40-60 words.

Exam Strategy

- Read the command word carefully: state, describe, explain, compare, calculate or evaluate.
- Use precise scientific terms and show the link between cause and effect.
- For data questions, quote relevant evidence and include units.

Notes & Corrections

Record one mistake, the correction and the biological principle.

LESSON 07 - CORE BIOLOGY

Nutrition in Plants

Concepts, functions, processes and model examples

Learning Objectives

- Describe autotrophic nutrition.
- State mineral requirements of plants.
- Explain root uptake of water and mineral ions.

Concept	Explanation	Example
Autotrophic nutrition	Organisms manufacture organic food from simple inorganic substances.	Green plants make glucose.
Nitrate ions	Mineral ions required to make amino acids and proteins.	Deficiency may cause poor growth.
Magnesium ions	Required to make chlorophyll.	Deficiency may cause yellow leaves.
Root hair cell	Specialized cell with a long extension for absorption.	Large surface area for water and mineral uptake.
Transpiration stream	Movement of water from roots through xylem to leaves.	Supplies water for photosynthesis and cooling.

Key Points

- Plants need light, carbon dioxide, water and mineral ions.
- Root hair cells increase the surface area for absorption.
- Water enters roots mainly by osmosis; mineral ions may enter by active transport.

Study Tip

Read each explanation, connect the structure to its function, and then create one new biological example using the same principle.

Extra Model Statements

1. Nitrate is needed for protein synthesis.
2. Magnesium is needed to make chlorophyll.
3. Root hairs contact water films around soil particles.
4. Xylem transports water and dissolved mineral ions upward.

Scientific Analysis Routine

Step 1: Identify the biological structure, process or variable.

Step 2: State the relevant scientific principle.

Step 3: Link structure to function or cause to effect.

Step 4: Support the explanation with a clear example or evidence.

Your Example

Write one accurate biological example and underline the key scientific term.

LESSON 07 - PRACTICE

Nutrition in Plants

Common misconceptions, guided practice and mastery check

Common error	Better scientific idea	Why?
Plants obtain all food from soil.	Plants manufacture sugars by photosynthesis; soil mainly supplies water and mineral ions.	Organic carbon comes mainly from carbon dioxide.
Mineral ions enter only by diffusion.	Some ions are absorbed by active transport.	Soil concentrations may be lower than inside root cells.

Guided Practice

1. What is autotrophic nutrition?
2. Why do plants need nitrate ions?
3. Why do plants need magnesium ions?
4. How are root hair cells adapted for absorption?
5. Which tissue transports water upward?
6. How does water usually enter root hair cells?
7. Which process may absorb mineral ions against a gradient?
8. Give one symptom of magnesium deficiency.

Mastery Challenge

- Write two original explanations demonstrating the main ideas from Nutrition in Plants.
- Use correct biological vocabulary and connect evidence to the conclusion.
- Check units, labels and scientific spellings before comparing with the answers.

Quick Answers

1. Making organic food from simple inorganic substances.
2. To synthesize amino acids and proteins.
3. To make chlorophyll.
4. They have a long extension and large surface area.
5. Xylem.
6. Osmosis.
7. Active transport.
8. Yellowing of leaves due to reduced chlorophyll.

Independent Practice

- A. Write three accurate statements using the main concepts from this lesson.
- B. Draw and label one relevant biological diagram or process flow where appropriate.
- C. Write one deliberately incorrect statement, then correct it and explain the scientific reason.
- D. Apply the lesson to a short real-life or laboratory context of 40-60 words.

Exam Strategy

- Read the command word carefully: state, describe, explain, compare, calculate or evaluate.
- Use precise scientific terms and show the link between cause and effect.
- For data questions, quote relevant evidence and include units.

Notes & Corrections

Record one mistake, the correction and the biological principle.

LESSON 08 - CORE BIOLOGY

Photosynthesis

Concepts, functions, processes and model examples

Learning Objectives

- Write the word equation for photosynthesis.
- Explain limiting factors of photosynthesis.
- Describe tests for starch in leaves.

Concept	Explanation	Example
Photosynthesis	Process by which plants use light energy to make glucose from carbon dioxide and water.	Occurs mainly in chloroplasts.
Chlorophyll	Green pigment that absorbs light energy.	Found in chloroplast membranes.
Limiting factor	A factor in shortest supply that restricts the rate.	Light, carbon dioxide or temperature.
Starch test	Leaf is decolorized and iodine is added.	Blue-black indicates starch.
Glucose use	Glucose can be respired, stored or used to build other substances.	Converted to starch or cellulose.

Key Points

- Word equation: carbon dioxide + water → glucose + oxygen, using light and chlorophyll.
- Rate increases with a limiting factor until another factor becomes limiting.
- Leaves require carbon dioxide, water, light and chlorophyll for normal photosynthesis.

Study Tip

Read each explanation, connect the structure to its function, and then create one new biological example using the same principle.

Extra Model Statements

1. Light energy is converted into chemical energy in glucose.
2. Starch is an insoluble storage carbohydrate.
3. Carbon dioxide enters leaves through stomata.
4. Oxygen is released as a by-product.

Scientific Analysis Routine

Step 1: Identify the biological structure, process or variable.

Step 2: State the relevant scientific principle.

Step 3: Link structure to function or cause to effect.

Step 4: Support the explanation with a clear example or evidence.

Your Example

Write one accurate biological example and underline the key scientific term.

LESSON 08 - PRACTICE

Photosynthesis

Common misconceptions, guided practice and mastery check

Common error	Better scientific idea	Why?
Plants photosynthesize only during the day and do not respire.	Plants respire all the time; photosynthesis occurs when enough light is available.	Both processes can occur simultaneously.
Oxygen is taken in to make glucose.	Carbon dioxide is the carbon source for glucose; oxygen is mainly released.	Gas exchange direction depends on net rates.

Guided Practice

1. Write the word equation for photosynthesis.
2. Where does photosynthesis mainly occur?
3. What is chlorophyll?
4. Name three limiting factors.
5. Why is a leaf boiled in alcohol during a starch test?
6. What color indicates starch?
7. How does carbon dioxide enter a leaf?
8. Give two uses of glucose in plants.

Mastery Challenge

- Write two original explanations demonstrating the main ideas from Photosynthesis.
- Use correct biological vocabulary and connect evidence to the conclusion.
- Check units, labels and scientific spellings before comparing with the answers.

Quick Answers

1. Carbon dioxide + water → glucose + oxygen.
2. In chloroplasts, especially in leaf cells.
3. A green pigment that absorbs light energy.
4. Light intensity, carbon dioxide concentration and temperature.
5. To remove chlorophyll so the iodine color change is visible.
6. Blue-black.
7. Through stomata by diffusion.
8. For example, respiration and storage as starch.

Independent Practice

- A. Write three accurate statements using the main concepts from this lesson.
- B. Draw and label one relevant biological diagram or process flow where appropriate.
- C. Write one deliberately incorrect statement, then correct it and explain the scientific reason.
- D. Apply the lesson to a short real-life or laboratory context of 40-60 words.

Exam Strategy

- Read the command word carefully: state, describe, explain, compare, calculate or evaluate.
- Use precise scientific terms and show the link between cause and effect.
- For data questions, quote relevant evidence and include units.

Notes & Corrections

Record one mistake, the correction and the biological principle.

LESSON 09 - CORE BIOLOGY

Mineral Nutrition in Plants

Concepts, functions, processes and model examples

Learning Objectives

- Relate major mineral ions to plant functions.
- Identify common deficiency symptoms.
- Explain the value and risks of fertilizers.

Concept	Explanation	Example
Nitrate	Needed to make amino acids and proteins.	Deficiency: stunted growth.
Magnesium	Central component of chlorophyll.	Deficiency: chlorosis or yellow leaves.
Phosphate	Needed in nucleic acids, ATP and membranes.	Deficiency can reduce root growth.
Potassium	Important in enzyme function and stomatal control.	Deficiency may weaken growth.
Fertilizer	Material added to soil to supply mineral nutrients.	NPK fertilizer.

Key Points

- Mineral nutrients are required in small amounts but are essential for healthy growth.
- Deficiency symptoms can overlap, so diagnosis should consider several signs.
- Excess fertilizer may leach into water and contribute to eutrophication.

Study Tip

Read each explanation, connect the structure to its function, and then create one new biological example using the same principle.

Extra Model Statements

1. NPK fertilizers supply nitrogen, phosphorus and potassium.
2. Magnesium deficiency reduces chlorophyll formation.
3. Nitrate deficiency reduces protein synthesis.
4. Plants absorb mineral ions from soil solution.

Scientific Analysis Routine

Step 1: Identify the biological structure, process or variable.

Step 2: State the relevant scientific principle.

Step 3: Link structure to function or cause to effect.

Step 4: Support the explanation with a clear example or evidence.

Your Example

Write one accurate biological example and underline the key scientific term.

LESSON 09 - PRACTICE

Mineral Nutrition in Plants

Common misconceptions, guided practice and mastery check

Common error	Better scientific idea	Why?
More fertilizer is always better.	Excess fertilizer can damage plants and pollute water.	Nutrients have optimum ranges.
Yellow leaves always mean lack of water.	Yellowing can have several causes, including mineral deficiency.	Symptoms require careful diagnosis.

Guided Practice

1. What does NPK stand for?
2. Which ion is needed for chlorophyll?
3. Which ion is strongly linked to amino acid production?
4. Name one possible effect of nitrate deficiency.
5. Why can fertilizer cause water pollution?
6. What is chlorosis?
7. How do plants absorb mineral ions?
8. Why can deficiency diagnosis be difficult?

Mastery Challenge

- Write two original explanations demonstrating the main ideas from Mineral Nutrition in Plants.
- Use correct biological vocabulary and connect evidence to the conclusion.
- Check units, labels and scientific spellings before comparing with the answers.

Quick Answers

1. Nitrogen, phosphorus and potassium.
2. Magnesium.
3. Nitrate.
4. Stunted growth.
5. Ions can leach into water and promote algal growth.
6. Yellowing caused by loss or lack of chlorophyll.
7. Through roots, often by active transport.
8. Different deficiencies can produce similar symptoms.

Independent Practice

- A. Write three accurate statements using the main concepts from this lesson.
- B. Draw and label one relevant biological diagram or process flow where appropriate.
- C. Write one deliberately incorrect statement, then correct it and explain the scientific reason.
- D. Apply the lesson to a short real-life or laboratory context of 40-60 words.

Exam Strategy

- Read the command word carefully: state, describe, explain, compare, calculate or evaluate.
- Use precise scientific terms and show the link between cause and effect.
- For data questions, quote relevant evidence and include units.

Notes & Corrections

Record one mistake, the correction and the biological principle.

LESSON 10 - CORE BIOLOGY

Human Nutrition and Digestion

Concepts, functions, processes and model examples

Learning Objectives

- Describe a balanced diet.
- Explain mechanical and chemical digestion.
- State functions of major digestive organs.

Concept	Explanation	Example
Balanced diet	A diet containing appropriate proportions of nutrients, fiber and water.	Includes carbohydrates, proteins, lipids, vitamins and minerals.
Digestion	Breakdown of large insoluble food molecules into small soluble molecules.	Proteins are digested to amino acids.
Amylase	Enzyme that digests starch to smaller sugars.	Acts in mouth and small intestine.
Protease	Enzyme that digests proteins.	Pepsin and trypsin are examples.
Bile	Alkaline fluid that emulsifies fats and helps neutralize stomach acid.	Made by liver, stored in gall bladder.

Key Points

- Mechanical digestion increases surface area; chemical digestion uses enzymes.
- Most absorption of digested nutrients occurs in the small intestine.
- Villi increase the surface area for absorption.

Study Tip

Read each explanation, connect the structure to its function, and then create one new biological example using the same principle.

Extra Model Statements

1. Teeth mechanically break food into smaller pieces.
2. The stomach provides acidic conditions for some proteases.
3. Bile emulsifies lipids into small droplets.
4. Villi have thin walls and a good blood supply.

Scientific Analysis Routine

Step 1: Identify the biological structure, process or variable.

Step 2: State the relevant scientific principle.

Step 3: Link structure to function or cause to effect.

Step 4: Support the explanation with a clear example or evidence.

Your Example

Write one accurate biological example and underline the key scientific term.

LESSON 10 - PRACTICE

Human Nutrition and Digestion

Common misconceptions, guided practice and mastery check

Common error	Better scientific idea	Why?
Bile is an enzyme.	Bile is not an enzyme; it emulsifies fat and helps neutralize acid.	It supports digestion without catalyzing hydrolysis.
Absorption occurs mainly in the stomach.	Most nutrient absorption occurs in the small intestine.	The stomach mainly stores, mixes and begins protein digestion.

Guided Practice

1. What is a balanced diet?
2. Define digestion.
3. What does amylase digest?
4. What is the role of bile?
5. Where is bile produced?
6. Where does most absorption occur?
7. How are villi adapted for absorption?
8. Distinguish mechanical from chemical digestion.

Mastery Challenge

- Write two original explanations demonstrating the main ideas from Human Nutrition and Digestion.
- Use correct biological vocabulary and connect evidence to the conclusion.
- Check units, labels and scientific spellings before comparing with the answers.

Quick Answers

1. A diet with suitable proportions of nutrients, fiber and water.
2. Breakdown of large insoluble food molecules into small soluble molecules.
3. Starch.
4. It emulsifies fats and helps neutralize acid.
5. The liver.
6. The small intestine.
7. Large surface area, thin surface and good blood supply.
8. Mechanical digestion physically breaks food; chemical digestion uses enzymes.

Independent Practice

- A. Write three accurate statements using the main concepts from this lesson.
- B. Draw and label one relevant biological diagram or process flow where appropriate.
- C. Write one deliberately incorrect statement, then correct it and explain the scientific reason.
- D. Apply the lesson to a short real-life or laboratory context of 40-60 words.

Exam Strategy

- Read the command word carefully: state, describe, explain, compare, calculate or evaluate.
- Use precise scientific terms and show the link between cause and effect.
- For data questions, quote relevant evidence and include units.

Notes & Corrections

Record one mistake, the correction and the biological principle.

LESSON 11 - CORE BIOLOGY

Respiration

Concepts, functions, processes and model examples

Learning Objectives

- Distinguish aerobic and anaerobic respiration.
- Write word equations for respiration.
- Explain why respiration is essential.

Concept	Explanation	Example
Aerobic respiration	Release of energy using oxygen.	Glucose + oxygen → carbon dioxide + water + energy.
Anaerobic respiration in muscles	Release of some energy without enough oxygen.	Glucose → lactic acid + energy.
Anaerobic respiration in yeast	Fermentation without oxygen.	Glucose → ethanol + carbon dioxide + energy.
ATP	Energy-transfer molecule produced in respiration.	Supports active transport and muscle contraction.
Oxygen debt	Extra oxygen needed after strenuous exercise to process accumulated lactate.	Breathing remains rapid after exercise.

Key Points

- Respiration occurs in all living cells.
- Aerobic respiration releases more usable energy per glucose molecule than anaerobic respiration.
- Anaerobic pathways differ between organisms.

Study Tip

Read each explanation, connect the structure to its function, and then create one new biological example using the same principle.

Extra Model Statements

1. Muscles may respire anaerobically during intense exercise.
2. Yeast fermentation produces ethanol and carbon dioxide.
3. Respiration provides energy for growth, active transport and movement.
4. Mitochondria are major sites of aerobic respiration.

Scientific Analysis Routine

Step 1: Identify the biological structure, process or variable.

Step 2: State the relevant scientific principle.

Step 3: Link structure to function or cause to effect.

Step 4: Support the explanation with a clear example or evidence.

Your Example

Write one accurate biological example and underline the key scientific term.

LESSON 11 - PRACTICE

Respiration

Common misconceptions, guided practice and mastery check

Common error	Better scientific idea	Why?
Respiration happens only in lungs.	Respiration is a cellular process; lungs exchange gases.	Gas exchange supplies oxygen for respiration.
Anaerobic respiration uses no glucose.	It uses glucose but not oxygen.	Less energy is released.

Guided Practice

1. Write the word equation for aerobic respiration.
2. Where does most aerobic respiration occur?
3. What is produced in anaerobic respiration in muscles?
4. What is produced in yeast fermentation?
5. Why is respiration important?
6. Which type releases more energy per glucose?
7. What is oxygen debt?
8. Give one use of energy released by respiration.

Mastery Challenge

- Write two original explanations demonstrating the main ideas from Respiration.
- Use correct biological vocabulary and connect evidence to the conclusion.
- Check units, labels and scientific spellings before comparing with the answers.

Quick Answers

1. Glucose + oxygen → carbon dioxide + water + energy.
2. Mainly in mitochondria.
3. Lactic acid and some energy.
4. Ethanol, carbon dioxide and some energy.
5. It supplies usable energy for cell processes.
6. Aerobic respiration.
7. Extra oxygen required after exercise to deal with accumulated lactate and restore normal conditions.
8. For example, active transport or muscle contraction.

Independent Practice

- A. Write three accurate statements using the main concepts from this lesson.
- B. Draw and label one relevant biological diagram or process flow where appropriate.
- C. Write one deliberately incorrect statement, then correct it and explain the scientific reason.
- D. Apply the lesson to a short real-life or laboratory context of 40-60 words.

Exam Strategy

- Read the command word carefully: state, describe, explain, compare, calculate or evaluate.
- Use precise scientific terms and show the link between cause and effect.
- For data questions, quote relevant evidence and include units.

Notes & Corrections

Record one mistake, the correction and the biological principle.

LESSON 12 - CORE BIOLOGY

Gas Exchange

Concepts, functions, processes and model examples

Learning Objectives

- Describe gas exchange surfaces.
- Explain ventilation in humans.
- Relate alveolar structure to efficient diffusion.

Concept	Explanation	Example
Alveolus	Tiny air sac in lungs where gas exchange occurs.	Oxygen diffuses into capillaries.
Ventilation	Movement of air into and out of the lungs.	Inhalation and exhalation.
Diffusion gradient	Difference in gas concentration driving diffusion.	Oxygen concentration is higher in alveolar air than deoxygenated blood.
Diaphragm	Sheet of muscle below lungs involved in ventilation.	Contracts and flattens during inhalation.
Cilia and mucus	Airway defenses that trap and remove particles.	Reduce entry of dust and pathogens.

Key Points

- Efficient gas exchange surfaces are thin, moist, have large area and good blood flow.
- Ventilation maintains steep concentration gradients.
- Smoking damages cilia, alveoli and cardiovascular health.

Study Tip

Read each explanation, connect the structure to its function, and then create one new biological example using the same principle.

Extra Model Statements

1. Alveoli provide a very large total surface area.
2. Capillary walls are one cell thick.
3. During inhalation, thoracic volume increases and pressure falls.
4. Oxygen binds to hemoglobin in red blood cells.

Scientific Analysis Routine

Step 1: Identify the biological structure, process or variable.

Step 2: State the relevant scientific principle.

Step 3: Link structure to function or cause to effect.

Step 4: Support the explanation with a clear example or evidence.

Your Example

Write one accurate biological example and underline the key scientific term.

LESSON 12 - PRACTICE

Gas Exchange

Common misconceptions, guided practice and mastery check

Common error	Better scientific idea	Why?
Air is pulled into lungs by a pump in the lung itself.	Pressure changes caused by respiratory muscles move air.	The diaphragm and intercostal muscles change thoracic volume.
Gas exchange is active transport.	Respiratory gases move mainly by diffusion.	No ATP is needed for diffusion itself.

Guided Practice

- Where does gas exchange occur in the lungs?
- Name four adaptations of alveoli.
- What happens to the diaphragm during inhalation?
- Why is a large surface area useful?
- How does ventilation help diffusion?
- What do cilia do?
- Which gas diffuses from blood into alveoli?
- Which protein carries most oxygen in blood?

Mastery Challenge

- Write two original explanations demonstrating the main ideas from Gas Exchange.
- Use correct biological vocabulary and connect evidence to the conclusion.
- Check units, labels and scientific spellings before comparing with the answers.

Quick Answers

- In the alveoli.
- Large area, thin walls, moist surface and good blood supply.
- It contracts and flattens.
- It increases the rate of gas exchange.
- It refreshes air and maintains steep concentration gradients.
- They move mucus and trapped particles toward the throat.
- Carbon dioxide.
- Hemoglobin.

Independent Practice

- Write three accurate statements using the main concepts from this lesson.
- Draw and label one relevant biological diagram or process flow where appropriate.
- Write one deliberately incorrect statement, then correct it and explain the scientific reason.
- Apply the lesson to a short real-life or laboratory context of 40-60 words.

Exam Strategy

- Read the command word carefully: state, describe, explain, compare, calculate or evaluate.
- Use precise scientific terms and show the link between cause and effect.
- For data questions, quote relevant evidence and include units.

Notes & Corrections

Record one mistake, the correction and the biological principle.

LESSON 13 - CORE BIOLOGY

Transport in Humans

Concepts, functions, processes and model examples

Learning Objectives

- Describe the composition of blood.
- Explain functions of plasma, red cells, white cells and platelets.
- Relate blood structure to transport.

Concept	Explanation	Example
Plasma	Liquid component transporting dissolved substances and heat.	Carries glucose, hormones, urea and carbon dioxide.
Red blood cell	Biconcave cell containing hemoglobin for oxygen transport.	Lacks nucleus when mature in mammals.
White blood cell	Cell involved in immune defense.	Phagocytes and lymphocytes.
Platelet	Cell fragment involved in blood clotting.	Helps form a clot after injury.
Hemoglobin	Protein that binds oxygen reversibly.	Forms oxyhemoglobin in oxygen-rich conditions.

Key Points

- Blood transports respiratory gases, nutrients, hormones, waste and heat.
- Red blood cells are adapted by their biconcave shape and high hemoglobin content.
- White blood cells defend against pathogens in different ways.

Study Tip

Read each explanation, connect the structure to its function, and then create one new biological example using the same principle.

Extra Model Statements

1. Plasma transports urea to the kidneys.
2. Platelets reduce blood loss after vessel damage.
3. Lymphocytes can produce antibodies.
4. Phagocytes engulf pathogens.

Scientific Analysis Routine

Step 1: Identify the biological structure, process or variable.

Step 2: State the relevant scientific principle.

Step 3: Link structure to function or cause to effect.

Step 4: Support the explanation with a clear example or evidence.

Your Example

Write one accurate biological example and underline the key scientific term.

LESSON 13 - PRACTICE

Transport in Humans

Common misconceptions, guided practice and mastery check

Common error	Better scientific idea	Why?
Plasma carries only water.	Plasma carries many dissolved substances and distributes heat.	It is a complex transport medium.
All white blood cells make antibodies.	Only some lymphocytes produce antibodies; other white cells have different roles.	Immune cells are specialized.

Guided Practice

1. Name four components of blood.
2. What does plasma transport?
3. How are red blood cells adapted for oxygen transport?
4. What is hemoglobin?
5. What is the role of platelets?
6. What do phagocytes do?
7. What can lymphocytes produce?
8. Why is the biconcave shape useful?

Mastery Challenge

- Write two original explanations demonstrating the main ideas from Transport in Humans.
- Use correct biological vocabulary and connect evidence to the conclusion.
- Check units, labels and scientific spellings before comparing with the answers.

Quick Answers

1. Plasma, red blood cells, white blood cells and platelets.
2. Dissolved nutrients, wastes, hormones, gases and heat.
3. Biconcave shape, no nucleus when mature, and much hemoglobin.
4. An oxygen-binding protein in red blood cells.
5. They help blood clot.
6. They engulf pathogens.
7. Antibodies.
8. It increases surface area and shortens diffusion distance.

Independent Practice

- A. Write three accurate statements using the main concepts from this lesson.
- B. Draw and label one relevant biological diagram or process flow where appropriate.
- C. Write one deliberately incorrect statement, then correct it and explain the scientific reason.
- D. Apply the lesson to a short real-life or laboratory context of 40-60 words.

Exam Strategy

- Read the command word carefully: state, describe, explain, compare, calculate or evaluate.
- Use precise scientific terms and show the link between cause and effect.
- For data questions, quote relevant evidence and include units.

Notes & Corrections

Record one mistake, the correction and the biological principle.

LESSON 14 - CORE BIOLOGY

The Heart and Blood Circulation

Concepts, functions, processes and model examples

Learning Objectives

- Describe heart structure and double circulation.
- Compare arteries, veins and capillaries.
- Explain the cardiac cycle in simple terms.

Concept	Explanation	Example
Atrium	Upper chamber receiving blood into the heart.	Right atrium receives blood from vena cava.
Ventricle	Lower chamber pumping blood out of the heart.	Left ventricle pumps to the body.
Valve	Structure preventing backflow.	Atrioventricular valves.
Artery	Vessel carrying blood away from the heart.	Aorta.
Vein	Vessel carrying blood toward the heart.	Vena cava.

Key Points

- Humans have a double circulatory system: pulmonary and systemic circuits.
- The left ventricle has a thicker wall because it pumps blood around the whole body.
- Capillaries have very thin walls for exchange.

Study Tip

Read each explanation, connect the structure to its function, and then create one new biological example using the same principle.

Extra Model Statements

1. Pulmonary artery carries deoxygenated blood to the lungs.
2. Pulmonary vein carries oxygenated blood to the heart.
3. Coronary arteries supply heart muscle.
4. Pulse is related to pressure waves generated by heart contraction.

Scientific Analysis Routine

Step 1: Identify the biological structure, process or variable.

Step 2: State the relevant scientific principle.

Step 3: Link structure to function or cause to effect.

Step 4: Support the explanation with a clear example or evidence.

Your Example

Write one accurate biological example and underline the key scientific term.

LESSON 14 - PRACTICE

The Heart and Blood Circulation

Common misconceptions, guided practice and mastery check

Common error	Better scientific idea	Why?
All arteries carry oxygenated blood.	Arteries are defined by direction away from the heart; the pulmonary artery carries deoxygenated blood.	Direction, not oxygen content, defines the vessel.
Veins have thick muscular walls like arteries.	Veins generally have thinner walls and valves.	Pressure is lower in veins.

Guided Practice

1. What is double circulation?
2. Why is the left ventricle wall thick?
3. What is the function of valves?
4. Which vessel carries blood from heart to lungs?
5. Which vessel returns oxygenated blood from lungs?
6. How are capillaries adapted for exchange?
7. What do coronary arteries supply?
8. Define an artery.

Mastery Challenge

- Write two original explanations demonstrating the main ideas from The Heart and Blood Circulation.
- Use correct biological vocabulary and connect evidence to the conclusion.
- Check units, labels and scientific spellings before comparing with the answers.

Quick Answers

1. Blood passes through the heart twice in one complete circuit.
2. It generates high pressure for systemic circulation.
3. They prevent backflow.
4. Pulmonary artery.
5. Pulmonary vein.
6. They have very thin walls and form extensive networks.
7. The heart muscle.
8. A vessel carrying blood away from the heart.

Independent Practice

- A. Write three accurate statements using the main concepts from this lesson.
- B. Draw and label one relevant biological diagram or process flow where appropriate.
- C. Write one deliberately incorrect statement, then correct it and explain the scientific reason.
- D. Apply the lesson to a short real-life or laboratory context of 40-60 words.

Exam Strategy

- Read the command word carefully: state, describe, explain, compare, calculate or evaluate.
- Use precise scientific terms and show the link between cause and effect.
- For data questions, quote relevant evidence and include units.

Notes & Corrections

Record one mistake, the correction and the biological principle.

LESSON 15 - CORE BIOLOGY

Transport in Plants

Concepts, functions, processes and model examples

Learning Objectives

- Describe xylem and phloem.
- Explain transpiration and translocation.
- Relate leaf and root structures to transport.

Concept	Explanation	Example
Xylem	Tissue transporting water and mineral ions upward and providing support.	Made largely of dead, lignified vessels.
Phloem	Living tissue transporting sucrose and amino acids between sources and sinks.	Translocation can occur in different directions.
Transpiration	Loss of water vapor from aerial plant parts, mainly through stomata.	Creates pull on xylem water column.
Stoma	Pore controlled by guard cells.	Allows gas exchange and water vapor loss.
Translocation	Movement of assimilates such as sucrose through phloem.	From leaf source to growing root sink.

Key Points

- Xylem transport is mainly upward; phloem translocation can be bidirectional in different sieve tubes.
- Transpiration is affected by temperature, humidity, wind and light.
- Guard cells regulate stomatal opening.

Study Tip

Read each explanation, connect the structure to its function, and then create one new biological example using the same principle.

Extra Model Statements

1. Lignin strengthens xylem vessels.
2. Water evaporates from mesophyll cell walls before diffusing out through stomata.
3. Sugars made in leaves can be transported to storage organs.
4. Root pressure and cohesion help water transport.

Scientific Analysis Routine

Step 1: Identify the biological structure, process or variable.

Step 2: State the relevant scientific principle.

Step 3: Link structure to function or cause to effect.

Step 4: Support the explanation with a clear example or evidence.

Your Example

Write one accurate biological example and underline the key scientific term.

LESSON 15 - PRACTICE

Transport in Plants

Common misconceptions, guided practice and mastery check

Common error	Better scientific idea	Why?
Phloem transports only downward.	Phloem moves assimilates from sources to sinks, which can be above or below each other.	Direction depends on source-sink relationship.
Transpiration is water loss from roots.	Transpiration is mainly water vapor loss from leaves and stems.	Root water uptake replaces the loss.

Guided Practice

1. What does xylem transport?
2. What does phloem transport?
3. Define transpiration.
4. What is translocation?
5. What is a stoma?
6. Name two factors that increase transpiration.
7. Why is xylem lignified?
8. Can phloem transport in both directions?

Mastery Challenge

- Write two original explanations demonstrating the main ideas from Transport in Plants.
- Use correct biological vocabulary and connect evidence to the conclusion.
- Check units, labels and scientific spellings before comparing with the answers.

Quick Answers

1. Water and mineral ions.
2. Mainly sucrose and amino acids.
3. Loss of water vapor from aerial plant surfaces, mainly through stomata.
4. Movement of assimilates through phloem.
5. A pore in the leaf epidermis controlled by guard cells.
6. For example, higher temperature and stronger wind.
7. For support and resistance to collapse.
8. Yes, depending on source-sink relationships.

Independent Practice

- A. Write three accurate statements using the main concepts from this lesson.
- B. Draw and label one relevant biological diagram or process flow where appropriate.
- C. Write one deliberately incorrect statement, then correct it and explain the scientific reason.
- D. Apply the lesson to a short real-life or laboratory context of 40-60 words.

Exam Strategy

- Read the command word carefully: state, describe, explain, compare, calculate or evaluate.
- Use precise scientific terms and show the link between cause and effect.
- For data questions, quote relevant evidence and include units.

Notes & Corrections

Record one mistake, the correction and the biological principle.

LESSON 16 - CORE BIOLOGY

Excretion

Concepts, functions, processes and model examples

Learning Objectives

- Define excretion and distinguish it from egestion.
- Describe major human excretory organs.
- Explain kidney function at a basic level.

Concept	Explanation	Example
Excretion	Removal of toxic materials and metabolic waste from an organism.	Removal of urea and carbon dioxide.
Kidney	Organ that filters blood and regulates water and ions.	Produces urine.
Urea	Nitrogenous waste produced from excess amino acids in the liver.	Excreted mainly by kidneys.
Nephron	Functional unit of the kidney.	Carries out filtration and selective reabsorption.
Urine	Solution containing urea, excess water and salts.	Stored in bladder before excretion.

Key Points

- Lungs excrete carbon dioxide; kidneys excrete urea, water and ions.
- Urea is formed in the liver from deamination of excess amino acids.
- Kidneys maintain internal conditions as well as remove waste.

Study Tip

Read each explanation, connect the structure to its function, and then create one new biological example using the same principle.

Extra Model Statements

1. Blood is filtered in nephrons.
2. Useful substances are selectively reabsorbed.
3. Urine flows through ureters to the bladder.
4. Sweat contains water and salts and small amounts of urea.

Scientific Analysis Routine

Step 1: Identify the biological structure, process or variable.

Step 2: State the relevant scientific principle.

Step 3: Link structure to function or cause to effect.

Step 4: Support the explanation with a clear example or evidence.

Your Example

Write one accurate biological example and underline the key scientific term.

LESSON 16 - PRACTICE

Excretion

Common misconceptions, guided practice and mastery check

Common error	Better scientific idea	Why?
Faeces are a metabolic waste product.	Faeces mainly contain undigested material and are removed by egestion.	Excretion concerns metabolic waste.
Kidneys make urea.	Urea is produced mainly in the liver and removed by the kidneys.	Kidneys filter and regulate blood composition.

Guided Practice

1. Define excretion.
2. Where is urea formed?
3. Which organs remove urea from blood?
4. What is a nephron?
5. Name three components of urine.
6. What do lungs excrete?
7. Distinguish excretion from egestion.
8. Where is urine stored?

Mastery Challenge

- Write two original explanations demonstrating the main ideas from Excretion.
- Use correct biological vocabulary and connect evidence to the conclusion.
- Check units, labels and scientific spellings before comparing with the answers.

Quick Answers

1. Removal of metabolic waste and toxic substances from an organism.
2. The liver.
3. The kidneys.
4. The functional unit of the kidney.
5. Urea, excess water and mineral ions.
6. Carbon dioxide and water vapor.
7. Excretion removes metabolic waste; egestion removes undigested food.
8. In the bladder.

Independent Practice

- A. Write three accurate statements using the main concepts from this lesson.
- B. Draw and label one relevant biological diagram or process flow where appropriate.
- C. Write one deliberately incorrect statement, then correct it and explain the scientific reason.
- D. Apply the lesson to a short real-life or laboratory context of 40-60 words.

Exam Strategy

- Read the command word carefully: state, describe, explain, compare, calculate or evaluate.
- Use precise scientific terms and show the link between cause and effect.
- For data questions, quote relevant evidence and include units.

Notes & Corrections

Record one mistake, the correction and the biological principle.

LESSON 17 - CORE BIOLOGY

Homeostasis

Concepts, functions, processes and model examples

Learning Objectives

- Define homeostasis.
- Explain temperature and blood glucose regulation.
- Describe negative feedback in simple terms.

Concept	Explanation	Example
Homeostasis	Maintenance of a relatively stable internal environment.	Keeping body temperature near a set range.
Negative feedback	Response that counteracts a change and restores conditions toward normal.	Insulin lowers high blood glucose.
Thermoregulation	Control of body temperature.	Sweating and changes in skin blood flow.
Insulin	Hormone that helps lower blood glucose.	Promotes uptake and storage of glucose.
Glucagon	Hormone that helps raise blood glucose.	Promotes breakdown of glycogen.

Key Points

- Homeostasis provides stable conditions for enzymes and cells.
- Negative feedback detects deviation and activates corrective responses.
- The hypothalamus is important in thermoregulation.

Study Tip

Read each explanation, connect the structure to its function, and then create one new biological example using the same principle.

Extra Model Statements

1. Sweating increases heat loss by evaporation.
2. Vasodilation increases blood flow near the skin surface.
3. Insulin is released when blood glucose is high.
4. Glucagon helps restore low blood glucose.

Scientific Analysis Routine

Step 1: Identify the biological structure, process or variable.

Step 2: State the relevant scientific principle.

Step 3: Link structure to function or cause to effect.

Step 4: Support the explanation with a clear example or evidence.

Your Example

Write one accurate biological example and underline the key scientific term.

LESSON 17 - PRACTICE

Homeostasis

Common misconceptions, guided practice and mastery check

Common error	Better scientific idea	Why?
Homeostasis keeps conditions completely constant.	Homeostasis maintains conditions within narrow ranges.	Biological systems fluctuate around target values.
Sweating cools because sweat is cold.	Cooling occurs mainly because evaporation uses thermal energy.	Evaporation is the key process.

Guided Practice

1. Define homeostasis.
2. What is negative feedback?
3. Name one response to high body temperature.
4. What happens to skin blood vessels when the body is hot?
5. Which hormone lowers blood glucose?
6. Which hormone raises blood glucose?
7. Why is stable temperature important?
8. Which brain region helps control body temperature?

Mastery Challenge

- Write two original explanations demonstrating the main ideas from Homeostasis.
- Use correct biological vocabulary and connect evidence to the conclusion.
- Check units, labels and scientific spellings before comparing with the answers.

Quick Answers

1. Maintenance of a relatively stable internal environment.
2. A control process that reverses a deviation from normal.
3. Sweating or vasodilation.
4. They dilate.
5. Insulin.
6. Glucagon.
7. Enzymes work best within suitable temperature ranges.
8. The hypothalamus.

Independent Practice

- A. Write three accurate statements using the main concepts from this lesson.
- B. Draw and label one relevant biological diagram or process flow where appropriate.
- C. Write one deliberately incorrect statement, then correct it and explain the scientific reason.
- D. Apply the lesson to a short real-life or laboratory context of 40-60 words.

Exam Strategy

- Read the command word carefully: state, describe, explain, compare, calculate or evaluate.
- Use precise scientific terms and show the link between cause and effect.
- For data questions, quote relevant evidence and include units.

Notes & Corrections

Record one mistake, the correction and the biological principle.

LESSON 18 - CORE BIOLOGY

Nervous System

Concepts, functions, processes and model examples

Learning Objectives

- Describe the organization of the nervous system.
- Explain reflex actions.
- Relate neuron structure to rapid communication.

Concept	Explanation	Example
Neuron	Specialized cell that transmits electrical impulses.	Sensory, relay and motor neurons.
Central nervous system	Brain and spinal cord.	Processes information and coordinates responses.
Receptor	Cell or organ that detects a stimulus.	Light receptors in the eye.
Effector	Muscle or gland that carries out a response.	Biceps muscle.
Reflex	Rapid, automatic response to a stimulus.	Withdrawal from a painful stimulus.

Key Points

- A reflex arc usually involves receptor → sensory neuron → relay neuron → motor neuron → effector.
- Synapses transfer signals between neurons using neurotransmitters.
- Reflexes protect the body and can occur before conscious awareness.

Study Tip

Read each explanation, connect the structure to its function, and then create one new biological example using the same principle.

Extra Model Statements

1. The pupil reflex controls light entering the eye.
2. Motor neurons carry impulses to muscles.
3. Sensory neurons carry impulses from receptors.
4. Myelin helps speed impulse conduction in many neurons.

Scientific Analysis Routine

Step 1: Identify the biological structure, process or variable.

Step 2: State the relevant scientific principle.

Step 3: Link structure to function or cause to effect.

Step 4: Support the explanation with a clear example or evidence.

Your Example

Write one accurate biological example and underline the key scientific term.

LESSON 18 - PRACTICE

Nervous System

Common misconceptions, guided practice and mastery check

Common error	Better scientific idea	Why?
Reflexes are always learned.	Many reflexes are innate and automatic.	Some conditioned responses are learned, but not all reflexes.
Neurons physically touch at synapses.	A tiny synaptic gap separates cells.	Chemical transmitters cross the gap.

Guided Practice

1. What is a neuron?
2. Name the parts of the central nervous system.
3. What is a receptor?
4. What is an effector?
5. Define a reflex action.
6. Write the order of a simple reflex arc.
7. What happens at a synapse?
8. What is one advantage of a reflex?

Mastery Challenge

- Write two original explanations demonstrating the main ideas from Nervous System.
- Use correct biological vocabulary and connect evidence to the conclusion.
- Check units, labels and scientific spellings before comparing with the answers.

Quick Answers

1. A specialized cell that transmits nerve impulses.
2. Brain and spinal cord.
3. A structure that detects a stimulus.
4. A muscle or gland that carries out a response.
5. A rapid automatic response to a stimulus.
6. Receptor -> sensory neuron -> relay neuron -> motor neuron -> effector.
7. Neurotransmitters carry the signal across the gap.
8. It provides a rapid protective response.

Independent Practice

- A. Write three accurate statements using the main concepts from this lesson.
- B. Draw and label one relevant biological diagram or process flow where appropriate.
- C. Write one deliberately incorrect statement, then correct it and explain the scientific reason.
- D. Apply the lesson to a short real-life or laboratory context of 40-60 words.

Exam Strategy

- Read the command word carefully: state, describe, explain, compare, calculate or evaluate.
- Use precise scientific terms and show the link between cause and effect.
- For data questions, quote relevant evidence and include units.

Notes & Corrections

Record one mistake, the correction and the biological principle.

LESSON 19 - CORE BIOLOGY

Hormones and Endocrine System

Concepts, functions, processes and model examples

Learning Objectives

- Define hormones and endocrine glands.
- Compare hormonal and nervous coordination.
- Give examples of major human hormones.

Concept	Explanation	Example
Hormone	Chemical messenger transported in blood to target cells.	Insulin.
Endocrine gland	Ductless gland that releases hormones into blood.	Pituitary, thyroid, pancreas.
Target cell	Cell with receptors for a particular hormone.	Liver cells respond to insulin and glucagon.
Adrenaline	Hormone preparing the body for action.	Increases heart rate and blood glucose availability.
Thyroxine	Hormone involved in metabolic rate.	Released by thyroid gland.

Key Points

- Hormonal responses are often slower but longer lasting than nervous responses.
- Only target cells with suitable receptors respond strongly to a hormone.
- Hormone levels are often controlled by feedback mechanisms.

Study Tip

Read each explanation, connect the structure to its function, and then create one new biological example using the same principle.

Extra Model Statements

1. The pancreas releases insulin and glucagon.
2. Adrenal glands release adrenaline.
3. The pituitary releases several coordinating hormones.
4. The thyroid releases thyroxine.

Scientific Analysis Routine

Step 1: Identify the biological structure, process or variable.

Step 2: State the relevant scientific principle.

Step 3: Link structure to function or cause to effect.

Step 4: Support the explanation with a clear example or evidence.

Your Example

Write one accurate biological example and underline the key scientific term.

LESSON 19 - PRACTICE

Hormones and Endocrine System

Common misconceptions, guided practice and mastery check

Common error	Better scientific idea	Why?
Hormones travel through nerves.	Hormones are transported mainly in blood.	Nerve impulses use neurons.
All cells respond to every hormone.	Only cells with appropriate receptors are target cells.	Receptor specificity controls response.

Guided Practice

1. Define a hormone.
2. What is an endocrine gland?
3. How are hormones transported?
4. Which gland releases adrenaline?
5. Which gland releases thyroxine?
6. Which organ releases insulin?
7. Give one difference between hormonal and nervous control.
8. What is a target cell?

Mastery Challenge

- Write two original explanations demonstrating the main ideas from Hormones and Endocrine System.
- Use correct biological vocabulary and connect evidence to the conclusion.
- Check units, labels and scientific spellings before comparing with the answers.

Quick Answers

1. A chemical messenger carried in blood to target cells.
2. A ductless gland that releases hormones into blood.
3. In the bloodstream.
4. Adrenal glands.
5. Thyroid gland.
6. The pancreas.
7. Hormonal control is generally slower and longer-lasting.
8. A cell with receptors for a particular hormone.

Independent Practice

- A. Write three accurate statements using the main concepts from this lesson.
- B. Draw and label one relevant biological diagram or process flow where appropriate.
- C. Write one deliberately incorrect statement, then correct it and explain the scientific reason.
- D. Apply the lesson to a short real-life or laboratory context of 40-60 words.

Exam Strategy

- Read the command word carefully: state, describe, explain, compare, calculate or evaluate.
- Use precise scientific terms and show the link between cause and effect.
- For data questions, quote relevant evidence and include units.

Notes & Corrections

Record one mistake, the correction and the biological principle.

LESSON 20 - CORE BIOLOGY

Human Reproduction

Concepts, functions, processes and model examples

Learning Objectives

- Describe male and female reproductive systems.
- Explain fertilization and early development.
- Describe the menstrual cycle at a basic level.

Concept	Explanation	Example
Gamete	Sex cell with a haploid chromosome set.	Sperm and egg.
Fertilization	Fusion of male and female gamete nuclei.	Forms a zygote.
Zygote	Diploid cell formed at fertilization.	Begins mitotic divisions.
Implantation	Attachment of early embryo to the uterine lining.	Occurs after several cell divisions.
Placenta	Organ for exchange between maternal and fetal blood supplies without direct mixing.	Transfers oxygen and nutrients.

Key Points

- Sperm are produced in testes; eggs develop in ovaries.
- Fertilization usually occurs in an oviduct.
- The placenta allows exchange of gases, nutrients and wastes.

Study Tip

Read each explanation, connect the structure to its function, and then create one new biological example using the same principle.

Extra Model Statements

1. Sperm have a flagellum for movement.
2. The egg contains nutrient-rich cytoplasm.
3. The uterus supports embryo and fetus development.
4. Hormones coordinate the menstrual cycle.

Scientific Analysis Routine

Step 1: Identify the biological structure, process or variable.

Step 2: State the relevant scientific principle.

Step 3: Link structure to function or cause to effect.

Step 4: Support the explanation with a clear example or evidence.

Your Example

Write one accurate biological example and underline the key scientific term.

LESSON 20 - PRACTICE

Human Reproduction

Common misconceptions, guided practice and mastery check

Common error	Better scientific idea	Why?
Fertilization occurs in the uterus.	Fertilization usually occurs in an oviduct.	Implantation later occurs in the uterus.
Maternal and fetal blood normally mix directly.	Exchange occurs across the placenta without normal direct mixing.	Separate circulations reduce risks.

Guided Practice

1. What is a gamete?
2. Where are sperm produced?
3. Where do eggs develop?
4. Define fertilization.
5. What is a zygote?
6. Where does fertilization usually occur?
7. What is implantation?
8. Give two functions of the placenta.

Mastery Challenge

- Write two original explanations demonstrating the main ideas from Human Reproduction.
- Use correct biological vocabulary and connect evidence to the conclusion.
- Check units, labels and scientific spellings before comparing with the answers.

Quick Answers

1. A haploid sex cell.
2. In the testes.
3. In the ovaries.
4. Fusion of male and female gamete nuclei.
5. The diploid cell formed after fertilization.
6. In an oviduct.
7. Attachment of an early embryo to the uterine lining.
8. Exchange of oxygen/nutrients and removal of fetal wastes, among other functions.

Independent Practice

- A. Write three accurate statements using the main concepts from this lesson.
- B. Draw and label one relevant biological diagram or process flow where appropriate.
- C. Write one deliberately incorrect statement, then correct it and explain the scientific reason.
- D. Apply the lesson to a short real-life or laboratory context of 40-60 words.

Exam Strategy

- Read the command word carefully: state, describe, explain, compare, calculate or evaluate.
- Use precise scientific terms and show the link between cause and effect.
- For data questions, quote relevant evidence and include units.

Notes & Corrections

Record one mistake, the correction and the biological principle.

LESSON 21 - CORE BIOLOGY

Plant Reproduction

Concepts, functions, processes and model examples

Learning Objectives

- Describe sexual reproduction in flowering plants.
- Explain pollination and fertilization.
- Describe seed and fruit formation.

Concept	Explanation	Example
Pollination	Transfer of pollen from anther to stigma of a flower of the same species.	May be by wind or animals.
Pollen grain	Structure carrying the male gamete nucleus.	Produced in anthers.
Ovule	Structure in ovary containing the female gamete.	Develops into a seed after fertilization.
Fertilization	Fusion of male and female gamete nuclei.	Occurs after pollen tube growth.
Seed dispersal	Movement of seeds away from parent plant.	Wind, animals, water or explosive mechanisms.

Key Points

- Pollination is not the same as fertilization.
- After fertilization, ovules usually develop into seeds and the ovary often develops into fruit.
- Cross-pollination can increase genetic variation.

Study Tip

Read each explanation, connect the structure to its function, and then create one new biological example using the same principle.

Extra Model Statements

1. Insect-pollinated flowers often have scented petals and nectar.
2. Wind-pollinated flowers often produce large quantities of light pollen.
3. Pollen tubes grow toward ovules.
4. Seeds contain an embryo and food reserve.

Scientific Analysis Routine

Step 1: Identify the biological structure, process or variable.

Step 2: State the relevant scientific principle.

Step 3: Link structure to function or cause to effect.

Step 4: Support the explanation with a clear example or evidence.

Your Example

Write one accurate biological example and underline the key scientific term.

LESSON 21 - PRACTICE

Plant Reproduction

Common misconceptions, guided practice and mastery check

Common error	Better scientific idea	Why?
Pollination is fusion of gametes.	Pollination is transfer of pollen; fertilization is fusion of nuclei.	They are separate stages.
All fruits are sweet and edible.	In Biology, a fruit is a mature ovary containing or associated with seeds.	Botanical definition is broader than everyday use.

Guided Practice

1. Define pollination.
2. Where is pollen produced?
3. What is an ovule?
4. Define fertilization.
5. What does an ovule usually become?
6. What does the ovary usually become?
7. Name two seed-dispersal methods.
8. Give one adaptation of wind-pollinated flowers.

Mastery Challenge

- Write two original explanations demonstrating the main ideas from Plant Reproduction.
- Use correct biological vocabulary and connect evidence to the conclusion.
- Check units, labels and scientific spellings before comparing with the answers.

Quick Answers

1. Transfer of pollen from anther to stigma.
2. In the anthers.
3. A structure in the ovary containing the female gamete.
4. Fusion of male and female gamete nuclei.
5. A seed.
6. A fruit.
7. For example, wind and animals.
8. For example, exposed anthers or large quantities of light pollen.

Independent Practice

- A. Write three accurate statements using the main concepts from this lesson.
- B. Draw and label one relevant biological diagram or process flow where appropriate.
- C. Write one deliberately incorrect statement, then correct it and explain the scientific reason.
- D. Apply the lesson to a short real-life or laboratory context of 40-60 words.

Exam Strategy

- Read the command word carefully: state, describe, explain, compare, calculate or evaluate.
- Use precise scientific terms and show the link between cause and effect.
- For data questions, quote relevant evidence and include units.

Notes & Corrections

Record one mistake, the correction and the biological principle.

LESSON 22 - CORE BIOLOGY

Growth and Development

Concepts, functions, processes and model examples

Learning Objectives

- Distinguish growth from development.
- Interpret simple growth curves.
- Explain factors influencing growth.

Concept	Explanation	Example
Growth	Permanent increase in size and dry mass.	Increase in height and biomass.
Development	Changes in complexity, form and function as an organism matures.	Puberty and organ maturation.
Cell division	Increase in cell number by mitosis.	Important in growth and repair.
Cell enlargement	Increase in cell size.	Common during plant growth.
Differentiation	Process by which cells become specialized.	Stem cells form specialized tissues.

Key Points

- Growth can be measured using mass, length, area or cell number depending on context.
- Development includes qualitative changes as well as growth.
- Genes and environment both influence growth patterns.

Study Tip

Read each explanation, connect the structure to its function, and then create one new biological example using the same principle.

Extra Model Statements

1. Plants grow at meristems.
2. Children show changes in body proportions during development.
3. Mitosis increases cell number.
4. Nutrition, hormones and health influence growth.

Scientific Analysis Routine

Step 1: Identify the biological structure, process or variable.

Step 2: State the relevant scientific principle.

Step 3: Link structure to function or cause to effect.

Step 4: Support the explanation with a clear example or evidence.

Your Example

Write one accurate biological example and underline the key scientific term.

LESSON 22 - PRACTICE

Growth and Development

Common misconceptions, guided practice and mastery check

Common error	Better scientific idea	Why?
Growth and development mean exactly the same thing.	Growth concerns measurable increase; development includes changes in form and function.	They are related but distinct concepts.
All body cells keep dividing throughout life.	Many cells become specialized and divide less often or not at all.	Division rate depends on cell type.

Guided Practice

1. Define growth.
2. Define development.
3. Which type of cell division increases body cell number?
4. What is differentiation?
5. Name two ways to measure growth.
6. Where do plants grow actively?
7. Give two environmental factors affecting growth.
8. Why is dry mass often useful in plant studies?

Mastery Challenge

- Write two original explanations demonstrating the main ideas from Growth and Development.
- Use correct biological vocabulary and connect evidence to the conclusion.
- Check units, labels and scientific spellings before comparing with the answers.

Quick Answers

1. Permanent increase in size and dry mass.
2. Progressive changes in form, complexity and function.
3. Mitosis.
4. The process by which cells become specialized.
5. For example, mass and length.
6. At meristems.
7. For example, nutrition and temperature.
8. It is less affected by temporary water content.

Independent Practice

- A. Write three accurate statements using the main concepts from this lesson.
- B. Draw and label one relevant biological diagram or process flow where appropriate.
- C. Write one deliberately incorrect statement, then correct it and explain the scientific reason.
- D. Apply the lesson to a short real-life or laboratory context of 40-60 words.

Exam Strategy

- Read the command word carefully: state, describe, explain, compare, calculate or evaluate.
- Use precise scientific terms and show the link between cause and effect.
- For data questions, quote relevant evidence and include units.

Notes & Corrections

Record one mistake, the correction and the biological principle.

LESSON 23 - CORE BIOLOGY

DNA, Genes and Chromosomes

Concepts, functions, processes and model examples

Learning Objectives

- Define DNA, gene and chromosome.
- Explain how genetic information is organized.
- Describe the relationship between genes and proteins.

Concept	Explanation	Example
DNA	Molecule that stores genetic information.	DNA forms chromosomes.
Gene	A length of DNA carrying information for a functional product, often a protein.	Gene influencing a particular enzyme.
Chromosome	Long DNA molecule associated with proteins.	Human somatic cells usually contain 46 chromosomes.
Allele	Alternative version of a gene.	Different alleles for a trait.
Genome	Complete genetic material of an organism.	All DNA in an organism.

Key Points

- DNA sequence carries information that cells use to make proteins and functional RNAs.
- Genes occur at specific positions on chromosomes.
- Different alleles can produce variation in characteristics.

Study Tip

Read each explanation, connect the structure to its function, and then create one new biological example using the same principle.

Extra Model Statements

1. Chromosomes are copied before cell division.
2. A mutation is a change in DNA sequence.
3. Proteins influence cell structure and function.
4. Inherited information passes from parents to offspring through gametes.

Scientific Analysis Routine

Step 1: Identify the biological structure, process or variable.

Step 2: State the relevant scientific principle.

Step 3: Link structure to function or cause to effect.

Step 4: Support the explanation with a clear example or evidence.

Your Example

Write one accurate biological example and underline the key scientific term.

LESSON 23 - PRACTICE

DNA, Genes and Chromosomes

Common misconceptions, guided practice and mastery check

Common error	Better scientific idea	Why?
A gene and a chromosome are the same thing.	A chromosome contains many genes.	Genes are sections of DNA.
All DNA directly codes for proteins.	Some DNA has regulatory or other non-protein-coding functions.	The genome contains multiple functional sequence types.

Guided Practice

1. What is DNA?
2. Define a gene.
3. What is a chromosome?
4. What is an allele?
5. What is a genome?
6. Where are genes located?
7. What is a mutation?
8. How can genes influence phenotype?

Mastery Challenge

- Write two original explanations demonstrating the main ideas from DNA, Genes and Chromosomes.
- Use correct biological vocabulary and connect evidence to the conclusion.
- Check units, labels and scientific spellings before comparing with the answers.

Quick Answers

1. The molecule that stores genetic information.
2. A length of DNA containing information for a functional product.
3. A long DNA molecule associated with proteins.
4. An alternative version of a gene.
5. The complete genetic material of an organism.
6. At specific positions on chromosomes.
7. A change in DNA sequence.
8. By influencing proteins and other functional products that affect cell function.

Independent Practice

- A. Write three accurate statements using the main concepts from this lesson.
- B. Draw and label one relevant biological diagram or process flow where appropriate.
- C. Write one deliberately incorrect statement, then correct it and explain the scientific reason.
- D. Apply the lesson to a short real-life or laboratory context of 40-60 words.

Exam Strategy

- Read the command word carefully: state, describe, explain, compare, calculate or evaluate.
- Use precise scientific terms and show the link between cause and effect.
- For data questions, quote relevant evidence and include units.

Notes & Corrections

Record one mistake, the correction and the biological principle.

LESSON 24 - CORE BIOLOGY

Cell Division: Mitosis and Meiosis

Concepts, functions, processes and model examples

Learning Objectives

- Compare mitosis and meiosis.
- Explain roles of mitosis in growth and repair.
- Explain how meiosis produces genetically varied gametes.

Concept	Explanation	Example
Mitosis	Nuclear division producing genetically similar daughter nuclei with the same chromosome number.	Growth and tissue repair.
Meiosis	Reduction division producing haploid cells and genetic variation.	Gamete production.
Diploid	Cell with two sets of chromosomes.	Most body cells.
Haploid	Cell with one set of chromosomes.	Gametes.
Chromosome replication	Copying DNA before division.	Produces sister chromatids.

Key Points

- Mitosis usually produces two genetically similar daughter cells.
- Meiosis involves two divisions and usually produces four haploid cells.
- Meiosis contributes to variation through chromosome assortment and crossing over.

Study Tip

Read each explanation, connect the structure to its function, and then create one new biological example using the same principle.

Extra Model Statements

1. Skin repair uses mitosis.
2. Sperm and eggs are produced through meiosis.
3. Fertilization restores the diploid chromosome number.
4. Cancer involves uncontrolled cell division.

Scientific Analysis Routine

Step 1: Identify the biological structure, process or variable.

Step 2: State the relevant scientific principle.

Step 3: Link structure to function or cause to effect.

Step 4: Support the explanation with a clear example or evidence.

Your Example

Write one accurate biological example and underline the key scientific term.

LESSON 24 - PRACTICE

Cell Division: Mitosis and Meiosis

Common misconceptions, guided practice and mastery check

Common error	Better scientific idea	Why?
Meiosis produces identical cells.	Meiosis produces genetically varied haploid cells.	Variation is a key outcome.
Mitosis halves chromosome number.	Mitosis maintains chromosome number.	Meiosis reduces it.

Guided Practice

1. What is mitosis?
2. What is meiosis?
3. How many daughter cells usually result from mitosis?
4. How many from meiosis?
5. What is a diploid cell?
6. What is a haploid cell?
7. Give two roles of mitosis.
8. Why does meiosis increase variation?

Mastery Challenge

- Write two original explanations demonstrating the main ideas from Cell Division: Mitosis and Meiosis.
- Use correct biological vocabulary and connect evidence to the conclusion.
- Check units, labels and scientific spellings before comparing with the answers.

Quick Answers

1. Cell division producing genetically similar cells with the same chromosome number.
2. Division producing haploid genetically varied cells.
3. Two.
4. Four.
5. A cell with two chromosome sets.
6. A cell with one chromosome set.
7. Growth and repair.
8. Because of processes such as independent assortment and crossing over.

Independent Practice

- A. Write three accurate statements using the main concepts from this lesson.
- B. Draw and label one relevant biological diagram or process flow where appropriate.
- C. Write one deliberately incorrect statement, then correct it and explain the scientific reason.
- D. Apply the lesson to a short real-life or laboratory context of 40-60 words.

Exam Strategy

- Read the command word carefully: state, describe, explain, compare, calculate or evaluate.
- Use precise scientific terms and show the link between cause and effect.
- For data questions, quote relevant evidence and include units.

Notes & Corrections

Record one mistake, the correction and the biological principle.

LESSON 25 - CORE BIOLOGY

Genetics and Inheritance

Concepts, functions, processes and model examples

Learning Objectives

- Use basic genetic terminology.
- Construct simple monohybrid crosses.
- Distinguish genotype and phenotype.

Concept	Explanation	Example
Genotype	Allele combination of an organism for a gene or genes.	AA, Aa or aa.
Phenotype	Observable characteristic influenced by genotype and environment.	Purple flower color.
Dominant allele	Allele expressed in a heterozygote in simple dominance.	A in Aa.
Recessive allele	Allele expressed phenotypically in simple dominance when no dominant allele is present.	aa phenotype.
Heterozygous	Having two different alleles for a gene.	Aa.

Key Points

- A Punnett square predicts expected probabilities, not guaranteed outcomes for a small family.
- Genotype influences phenotype, but environment can also affect many traits.
- Some inheritance patterns are more complex than simple dominance.

Study Tip

Read each explanation, connect the structure to its function, and then create one new biological example using the same principle.

Extra Model Statements

1. Aa x Aa can give genotype ratio 1 AA : 2 Aa : 1 aa in simple Mendelian inheritance.
2. A test cross can help infer an unknown dominant phenotype genotype.
3. Gametes contain one allele for each gene.
4. Fertilization combines alleles from two parents.

Scientific Analysis Routine

Step 1: Identify the biological structure, process or variable.

Step 2: State the relevant scientific principle.

Step 3: Link structure to function or cause to effect.

Step 4: Support the explanation with a clear example or evidence.

Your Example

Write one accurate biological example and underline the key scientific term.

LESSON 25 - PRACTICE

Genetics and Inheritance

Common misconceptions, guided practice and mastery check

Common error	Better scientific idea	Why?
A 3:1 ratio must occur in every four offspring.	Genetic ratios describe probabilities over many offspring.	Small samples can differ by chance.
Dominant means more common.	Dominant describes expression in a heterozygote, not population frequency.	A dominant allele can be rare.

Guided Practice

1. Define genotype.
2. Define phenotype.
3. What is a dominant allele?
4. What is a recessive allele?
5. What does heterozygous mean?
6. What alleles can an Aa parent place in gametes?
7. What is the expected genotype ratio from Aa x Aa?
8. Why may observed ratios differ from expected ratios?

Mastery Challenge

- Write two original explanations demonstrating the main ideas from Genetics and Inheritance.
- Use correct biological vocabulary and connect evidence to the conclusion.
- Check units, labels and scientific spellings before comparing with the answers.

Quick Answers

1. An organism's allele combination.
2. An observable characteristic.
3. An allele expressed in a heterozygote in simple dominance.
4. An allele masked by a dominant allele in a heterozygote.
5. Having two different alleles.
6. A or a.
7. 1 AA : 2 Aa : 1 aa.
8. Chance, especially in small samples.

Independent Practice

- A. Write three accurate statements using the main concepts from this lesson.
- B. Draw and label one relevant biological diagram or process flow where appropriate.
- C. Write one deliberately incorrect statement, then correct it and explain the scientific reason.
- D. Apply the lesson to a short real-life or laboratory context of 40-60 words.

Exam Strategy

- Read the command word carefully: state, describe, explain, compare, calculate or evaluate.
- Use precise scientific terms and show the link between cause and effect.
- For data questions, quote relevant evidence and include units.

Notes & Corrections

Record one mistake, the correction and the biological principle.

LESSON 26 - CORE BIOLOGY

Variation and Natural Selection

Concepts, functions, processes and model examples

Learning Objectives

- Distinguish continuous and discontinuous variation.
- Explain sources of variation.
- Describe natural selection.

Concept	Explanation	Example
Variation	Differences among individuals of the same species.	Height, blood group, color.
Continuous variation	Shows a range of values with many intermediates.	Human height.
Discontinuous variation	Falls into distinct categories.	ABO blood group.
Natural selection	Differential survival and reproduction of individuals with heritable variation.	Adaptations become more common over generations.
Selection pressure	Environmental factor affecting survival or reproductive success.	Predation, disease, drought.

Key Points

- Variation may result from genes, environment, or both.
- Natural selection acts on existing heritable variation.
- Evolution occurs across generations, not because individuals deliberately change.

Study Tip

Read each explanation, connect the structure to its function, and then create one new biological example using the same principle.

Extra Model Statements

1. Antibiotic resistance can spread when resistant bacteria survive treatment.
2. Camouflage can reduce predation.
3. Drought may favor plants with water-conserving traits.
4. Mutation provides new genetic variation.

Scientific Analysis Routine

Step 1: Identify the biological structure, process or variable.

Step 2: State the relevant scientific principle.

Step 3: Link structure to function or cause to effect.

Step 4: Support the explanation with a clear example or evidence.

Your Example

Write one accurate biological example and underline the key scientific term.

LESSON 26 - PRACTICE

Variation and Natural Selection

Common misconceptions, guided practice and mastery check

Common error	Better scientific idea	Why?
Organisms evolve because they need to.	Natural selection favors heritable variants that already differ in survival and reproduction.	Need does not direct mutations.
Natural selection changes an individual during life.	Selection changes allele frequencies in populations across generations.	Individuals are selected; populations evolve.

Guided Practice

1. Define variation.
2. Give one example of continuous variation.
3. Give one example of discontinuous variation.
4. Name two sources of variation.
5. Define natural selection.
6. What is a selection pressure?
7. How can antibiotic resistance spread?
8. Do individuals evolve during their lifetime?

Mastery Challenge

- Write two original explanations demonstrating the main ideas from Variation and Natural Selection.
- Use correct biological vocabulary and connect evidence to the conclusion.
- Check units, labels and scientific spellings before comparing with the answers.

Quick Answers

1. Differences among individuals of the same species.
2. For example, height.
3. For example, ABO blood group.
4. Genes and environment.
5. Differential survival and reproduction due to heritable differences.
6. An environmental factor affecting survival or reproduction.
7. Resistant bacteria survive and reproduce more successfully under antibiotic exposure.
8. No; populations evolve across generations.

Independent Practice

- A. Write three accurate statements using the main concepts from this lesson.
- B. Draw and label one relevant biological diagram or process flow where appropriate.
- C. Write one deliberately incorrect statement, then correct it and explain the scientific reason.
- D. Apply the lesson to a short real-life or laboratory context of 40-60 words.

Exam Strategy

- Read the command word carefully: state, describe, explain, compare, calculate or evaluate.
- Use precise scientific terms and show the link between cause and effect.
- For data questions, quote relevant evidence and include units.

Notes & Corrections

Record one mistake, the correction and the biological principle.

LESSON 27 - CORE BIOLOGY

Evolution

Concepts, functions, processes and model examples

Learning Objectives

- Define evolution in population terms.
- Explain evidence from fossils and molecular biology.
- Relate speciation to reproductive isolation.

Concept	Explanation	Example
Evolution	Change in heritable characteristics or allele frequencies of populations over generations.	Long-term population change.
Fossil	Preserved remains or traces of past organisms.	Bones, shells or impressions.
Common ancestry	Idea that related species descend from shared ancestral populations.	Supported by DNA similarities.
Speciation	Formation of new species.	Often involves reproductive isolation.
Adaptation	Heritable feature that improves survival or reproduction in a particular environment.	Thick fur in cold climates.

Key Points

- Evolutionary relationships are inferred from multiple lines of evidence.
- Fossil records are incomplete but can show change over geological time.
- Speciation often requires reduced gene flow between populations.

Study Tip

Read each explanation, connect the structure to its function, and then create one new biological example using the same principle.

Extra Model Statements

1. Similar DNA sequences can indicate close evolutionary relationships.
2. Geographical isolation can contribute to divergence.
3. Natural selection can adapt populations to local conditions.
4. Homologous structures can reflect common ancestry.

Scientific Analysis Routine

Step 1: Identify the biological structure, process or variable.

Step 2: State the relevant scientific principle.

Step 3: Link structure to function or cause to effect.

Step 4: Support the explanation with a clear example or evidence.

Your Example

Write one accurate biological example and underline the key scientific term.

LESSON 27 - PRACTICE

Evolution

Common misconceptions, guided practice and mastery check

Common error	Better scientific idea	Why?
Humans evolved from modern monkeys.	Humans and modern monkeys share common ancestors; neither evolved directly from the modern form of the other.	Evolution is branching, not a straight ladder.
Evolution always leads to greater complexity.	Evolution favors traits suited to current environments, not a predetermined direction.	Simplification can also be adaptive.

Guided Practice

1. Define evolution.
2. What is a fossil?
3. Give one molecular line of evolutionary evidence.
4. What is speciation?
5. What is reproductive isolation?
6. How can geographic isolation promote speciation?
7. What is an adaptation?
8. Why is the fossil record incomplete?

Mastery Challenge

- Write two original explanations demonstrating the main ideas from Evolution.
- Use correct biological vocabulary and connect evidence to the conclusion.
- Check units, labels and scientific spellings before comparing with the answers.

Quick Answers

1. Change in heritable population characteristics across generations.
2. Preserved remains or traces of past organisms.
3. DNA or protein sequence similarity.
4. Formation of new species.
5. Barriers preventing gene flow between populations.
6. Separated populations can diverge genetically under different pressures.
7. A heritable feature increasing survival or reproduction in an environment.
8. Many organisms do not fossilize and geological processes destroy or hide fossils.

Independent Practice

- A. Write three accurate statements using the main concepts from this lesson.
- B. Draw and label one relevant biological diagram or process flow where appropriate.
- C. Write one deliberately incorrect statement, then correct it and explain the scientific reason.
- D. Apply the lesson to a short real-life or laboratory context of 40-60 words.

Exam Strategy

- Read the command word carefully: state, describe, explain, compare, calculate or evaluate.
- Use precise scientific terms and show the link between cause and effect.
- For data questions, quote relevant evidence and include units.

Notes & Corrections

Record one mistake, the correction and the biological principle.

LESSON 28 - CORE BIOLOGY

Ecology and Ecosystems

Concepts, functions, processes and model examples

Learning Objectives

- Define ecological levels of organization.
- Describe biotic and abiotic factors.
- Explain interactions within ecosystems.

Concept	Explanation	Example
Population	Members of one species living in an area.	A population of baobab trees.
Community	All populations of different species in an area.	Plants, animals and microbes in a forest.
Ecosystem	Community plus its physical environment and interactions.	A lake ecosystem.
Biotic factor	Living influence on organisms.	Competition or predation.
Abiotic factor	Non-living environmental influence.	Light, temperature, water, pH.

Key Points

- Ecology studies relationships among organisms and between organisms and the environment.
- Population size is influenced by births, deaths, immigration and emigration.
- Limiting factors can constrain population growth.

Study Tip

Read each explanation, connect the structure to its function, and then create one new biological example using the same principle.

Extra Model Statements

1. Low rainfall can limit plant growth.
2. Predators influence prey populations.
3. Competition occurs when organisms require the same limited resource.
4. Decomposers recycle nutrients.

Scientific Analysis Routine

Step 1: Identify the biological structure, process or variable.

Step 2: State the relevant scientific principle.

Step 3: Link structure to function or cause to effect.

Step 4: Support the explanation with a clear example or evidence.

Your Example

Write one accurate biological example and underline the key scientific term.

LESSON 28 - PRACTICE

Ecology and Ecosystems

Common misconceptions, guided practice and mastery check

Common error	Better scientific idea	Why?
An ecosystem contains only living organisms.	An ecosystem includes both the community and physical environment.	Abiotic factors are essential components.
A habitat and a niche are the same.	A habitat is where an organism lives; a niche describes its role and resource use.	The concepts are related but different.

Guided Practice

1. Define population.
2. Define community.
3. Define ecosystem.
4. Give two abiotic factors.
5. Give two biotic factors.
6. What is competition?
7. What is a habitat?
8. What is a niche?

Mastery Challenge

- Write two original explanations demonstrating the main ideas from Ecology and Ecosystems.
- Use correct biological vocabulary and connect evidence to the conclusion.
- Check units, labels and scientific spellings before comparing with the answers.

Quick Answers

1. Members of one species living in an area.
2. All populations of different species in an area.
3. A community interacting with its physical environment.
4. For example, temperature and water.
5. For example, competition and predation.
6. Interaction when organisms require the same limited resource.
7. The place where an organism lives.
8. Its ecological role and pattern of resource use.

Independent Practice

- A. Write three accurate statements using the main concepts from this lesson.
- B. Draw and label one relevant biological diagram or process flow where appropriate.
- C. Write one deliberately incorrect statement, then correct it and explain the scientific reason.
- D. Apply the lesson to a short real-life or laboratory context of 40-60 words.

Exam Strategy

- Read the command word carefully: state, describe, explain, compare, calculate or evaluate.
- Use precise scientific terms and show the link between cause and effect.
- For data questions, quote relevant evidence and include units.

Notes & Corrections

Record one mistake, the correction and the biological principle.

LESSON 29 - CORE BIOLOGY

Food Chains, Food Webs and Nutrient Cycles

Concepts, functions, processes and model examples

Learning Objectives

- Construct food chains and food webs.
- Explain trophic levels and energy loss.
- Describe carbon and nitrogen cycling in outline.

Concept	Explanation	Example
Producer	Organism that makes organic food from inorganic materials.	Green plant.
Consumer	Organism that obtains energy by feeding on other organisms.	Herbivore or carnivore.
Decomposer	Organism that breaks down dead material and waste.	Many fungi and bacteria.
Trophic level	Feeding position in a food chain.	Producer is first trophic level.
Food web	Network of interconnected food chains.	Represents multiple feeding relationships.

Key Points

- Energy decreases at successive trophic levels because some is lost as heat, movement and waste.
- Matter is recycled through ecosystems; energy flows through and is ultimately lost as heat.
- Decomposers return mineral nutrients to the environment.

Study Tip

Read each explanation, connect the structure to its function, and then create one new biological example using the same principle.

Extra Model Statements

1. Grass → grasshopper → frog → snake.
2. Plants remove carbon dioxide during photosynthesis.
3. Respiration returns carbon dioxide to the atmosphere.
4. Nitrogen-fixing bacteria convert atmospheric nitrogen into usable forms.

Scientific Analysis Routine

Step 1: Identify the biological structure, process or variable.

Step 2: State the relevant scientific principle.

Step 3: Link structure to function or cause to effect.

Step 4: Support the explanation with a clear example or evidence.

Your Example

Write one accurate biological example and underline the key scientific term.

LESSON 29 - PRACTICE

Food Chains, Food Webs and Nutrient Cycles

Common misconceptions, guided practice and mastery check

Common error	Better scientific idea	Why?
Energy is recycled in ecosystems.	Matter cycles, but energy flows and is lost as heat.	Continuous energy input is needed.
Arrows in food chains show which organism is larger.	Arrows show direction of energy transfer.	They point from food to consumer.

Guided Practice

1. What is a producer?
2. What is a decomposer?
3. What do arrows in a food chain show?
4. Why does energy decrease at higher trophic levels?
5. What is a food web?
6. How does photosynthesis affect carbon dioxide?
7. How does respiration affect carbon dioxide?
8. What is nitrogen fixation?

Mastery Challenge

- Write two original explanations demonstrating the main ideas from Food Chains, Food Webs and Nutrient Cycles.
- Use correct biological vocabulary and connect evidence to the conclusion.
- Check units, labels and scientific spellings before comparing with the answers.

Quick Answers

1. An organism that makes organic food, usually by photosynthesis.
2. An organism that breaks down dead organic material and waste.
3. Direction of energy transfer.
4. Energy is lost as heat, movement, waste and uneaten material.
5. A network of interconnected food chains.
6. It removes carbon dioxide from the environment.
7. It releases carbon dioxide.
8. Conversion of atmospheric nitrogen into biologically usable nitrogen compounds.

Independent Practice

- A. Write three accurate statements using the main concepts from this lesson.
- B. Draw and label one relevant biological diagram or process flow where appropriate.
- C. Write one deliberately incorrect statement, then correct it and explain the scientific reason.
- D. Apply the lesson to a short real-life or laboratory context of 40-60 words.

Exam Strategy

- Read the command word carefully: state, describe, explain, compare, calculate or evaluate.
- Use precise scientific terms and show the link between cause and effect.
- For data questions, quote relevant evidence and include units.

Notes & Corrections

Record one mistake, the correction and the biological principle.

LESSON 30 - CORE BIOLOGY

Human Impact, Conservation and Biology Review

Concepts, functions, processes and model examples

Learning Objectives

- Describe major human impacts on ecosystems.
- Explain conservation strategies.
- Apply Biology concepts to unfamiliar problems.

Concept	Explanation	Example
Biodiversity	Variety of life at genetic, species and ecosystem levels.	High species richness in tropical forests.
Conservation	Protection and sustainable management of species, habitats and resources.	Protected areas and restoration.
Deforestation	Removal of forest cover.	Can reduce habitats and carbon storage.
Pollution	Introduction of harmful substances or energy into the environment.	Air, water and soil pollution.
Sustainable use	Use of resources at rates that allow long-term availability.	Managed fisheries and forestry.

Key Points

- Conservation aims to maintain biodiversity and ecosystem function while considering human needs.
- Habitat loss, overexploitation, pollution, invasive species and climate change can reduce biodiversity.
- Effective conservation often combines protected areas, laws, restoration, monitoring and community participation.

Study Tip

Read each explanation, connect the structure to its function, and then create one new biological example using the same principle.

Extra Model Statements

1. Reforestation can restore habitat and carbon storage.
2. Captive breeding may support endangered species recovery.
3. Wastewater treatment reduces nutrient pollution.
4. Sustainable harvest limits can protect populations from overexploitation.

Scientific Analysis Routine

Step 1: Identify the biological structure, process or variable.

Step 2: State the relevant scientific principle.

Step 3: Link structure to function or cause to effect.

Step 4: Support the explanation with a clear example or evidence.

Your Example

Write one accurate biological example and underline the key scientific term.

LESSON 30 - PRACTICE

Human Impact, Conservation and Biology Review

Common misconceptions, guided practice and mastery check

Common error	Better scientific idea	Why?
Conservation means never using natural resources.	Conservation can include sustainable use as well as strict protection.	The goal is long-term ecological viability.
One conservation method works everywhere.	Strategies must match local ecology, threats and social conditions.	Context matters.

Guided Practice

1. Define biodiversity.
2. Define conservation.
3. Give three major threats to biodiversity.
4. What is sustainable use?
5. How can deforestation affect ecosystems?
6. Give one method of species conservation.
7. Give one method of habitat conservation.
8. Why is monitoring important in conservation?

Mastery Challenge

- Write two original explanations demonstrating the main ideas from Human Impact, Conservation and Biology Review.
- Use correct biological vocabulary and connect evidence to the conclusion.
- Check units, labels and scientific spellings before comparing with the answers.

Quick Answers

1. Variety of life at genetic, species and ecosystem levels.
2. Protection and sustainable management of biodiversity and ecosystems.
3. For example, habitat loss, pollution and overexploitation.
4. Using resources at rates that allow long-term availability.
5. It removes habitat, can reduce biodiversity and changes water/carbon cycles.
6. For example, captive breeding.
7. For example, protected areas or habitat restoration.
8. It shows whether populations and habitats are improving or declining.

Independent Practice

- A. Write three accurate statements using the main concepts from this lesson.
- B. Draw and label one relevant biological diagram or process flow where appropriate.
- C. Write one deliberately incorrect statement, then correct it and explain the scientific reason.
- D. Apply the lesson to a short real-life or laboratory context of 40-60 words.

Exam Strategy

- Read the command word carefully: state, describe, explain, compare, calculate or evaluate.
- Use precise scientific terms and show the link between cause and effect.
- For data questions, quote relevant evidence and include units.

Notes & Corrections

Record one mistake, the correction and the biological principle.

REVISION 01

Review Test 1 - Foundations & Cells

Mixed practice - complete without looking back at the lessons.

1. Introduction to Biology and Characteristics of Living Things	What is Biology?
2. Cell Structure and Cell Organization	Define a cell.
3. Plant and Animal Cells	Name three structures present in plant cells but absent from typical animal cells.
4. Cell Membrane and Transport	Define diffusion.
5. Introduction to Biology and Characteristics of Living Things	Name the seven characteristics of living things.
6. Cell Structure and Cell Organization	Name two structures found in most animal cells.
7. Plant and Animal Cells	What is the cell wall made of?
8. Cell Membrane and Transport	Define osmosis.
9. Introduction to Biology and Characteristics of Living Things	Give one example of sensitivity.
10. Cell Structure and Cell Organization	What is the function of mitochondria?
11. Plant and Animal Cells	What is the function of chloroplasts?
12. Cell Membrane and Transport	Which transport process needs energy?

Writing Task

Write a short paragraph of 80-100 words using at least four biological concepts reviewed on this page. Underline the four target scientific terms.

Self-Assessment

- Score 10-12: Strong mastery.
- Score 7-9: Good progress; review minor errors.
- Score 4-6: Revisit the related lessons.
- Score 0-3: Study the lesson explanations before retrying.

Writing Task Response

REVISION 02

Review Test 2 - Molecules, Enzymes & Plants

Mixed practice - complete without looking back at the lessons.

1. Biological Molecules	Name the building blocks of proteins.
2. Enzymes	Define an enzyme.
3. Nutrition in Plants	What is autotrophic nutrition?
4. Photosynthesis	Write the word equation for photosynthesis.
5. Mineral Nutrition in Plants	What does NPK stand for?
6. Biological Molecules	Give two functions of lipids.
7. Enzymes	What is a substrate?
8. Nutrition in Plants	Why do plants need nitrate ions?
9. Photosynthesis	Where does photosynthesis mainly occur?
10. Mineral Nutrition in Plants	Which ion is needed for chlorophyll?
11. Biological Molecules	Which reagent tests for starch?
12. Enzymes	What happens at the active site?

Writing Task

Write a short paragraph of 80-100 words using at least four biological concepts reviewed on this page. Underline the four target scientific terms.

Self-Assessment

- Score 10-12: Strong mastery.
- Score 7-9: Good progress; review minor errors.
- Score 4-6: Revisit the related lessons.
- Score 0-3: Study the lesson explanations before retrying.

Writing Task Response

REVISION 03

Review Test 3 - Digestion, Respiration & Gas Exchange

Mixed practice - complete without looking back at the lessons.

1. Human Nutrition and Digestion	What is a balanced diet?
2. Respiration	Write the word equation for aerobic respiration.
3. Gas Exchange	Where does gas exchange occur in the lungs?
4. Human Nutrition and Digestion	Define digestion.
5. Respiration	Where does most aerobic respiration occur?
6. Gas Exchange	Name four adaptations of alveoli.
7. Human Nutrition and Digestion	What does amylase digest?
8. Respiration	What is produced in anaerobic respiration in muscles?
9. Gas Exchange	What happens to the diaphragm during inhalation?
10. Human Nutrition and Digestion	What is the role of bile?
11. Respiration	What is produced in yeast fermentation?
12. Gas Exchange	Why is a large surface area useful?

Writing Task

Write a short paragraph of 80-100 words using at least four biological concepts reviewed on this page. Underline the four target scientific terms.

Self-Assessment

- Score 10-12: Strong mastery.
- Score 7-9: Good progress; review minor errors.
- Score 4-6: Revisit the related lessons.
- Score 0-3: Study the lesson explanations before retrying.

Writing Task Response

REVISION 04

Review Test 4 - Transport Systems

Mixed practice - complete without looking back at the lessons.

1. Transport in Humans	Name four components of blood.
2. The Heart and Blood Circulation	What is double circulation?
3. Transport in Plants	What does xylem transport?
4. Transport in Humans	What does plasma transport?
5. The Heart and Blood Circulation	Why is the left ventricle wall thick?
6. Transport in Plants	What does phloem transport?
7. Transport in Humans	How are red blood cells adapted for oxygen transport?
8. The Heart and Blood Circulation	What is the function of valves?
9. Transport in Plants	Define transpiration.
10. Transport in Humans	What is hemoglobin?
11. The Heart and Blood Circulation	Which vessel carries blood from heart to lungs?
12. Transport in Plants	What is translocation?

Writing Task

Write a short paragraph of 80-100 words using at least four biological concepts reviewed on this page. Underline the four target scientific terms.

Self-Assessment

- Score 10-12: Strong mastery.
- Score 7-9: Good progress; review minor errors.
- Score 4-6: Revisit the related lessons.
- Score 0-3: Study the lesson explanations before retrying.

Writing Task Response

REVISION 05

Review Test 5 - Excretion & Coordination

Mixed practice - complete without looking back at the lessons.

1. Excretion	Define excretion.
2. Homeostasis	Define homeostasis.
3. Nervous System	What is a neuron?
4. Excretion	Where is urea formed?
5. Homeostasis	What is negative feedback?
6. Nervous System	Name the parts of the central nervous system.
7. Excretion	Which organs remove urea from blood?
8. Homeostasis	Name one response to high body temperature.
9. Nervous System	What is a receptor?
10. Excretion	What is a nephron?
11. Homeostasis	What happens to skin blood vessels when the body is hot?
12. Nervous System	What is an effector?

Writing Task

Write a short paragraph of 80-100 words using at least four biological concepts reviewed on this page. Underline the four target scientific terms.

Self-Assessment

- Score 10-12: Strong mastery.
- Score 7-9: Good progress; review minor errors.
- Score 4-6: Revisit the related lessons.
- Score 0-3: Study the lesson explanations before retrying.

Writing Task Response

REVISION 06

Review Test 6 - Reproduction, Growth & DNA

Mixed practice - complete without looking back at the lessons.

1. Hormones and Endocrine System	Define a hormone.
2. Human Reproduction	What is a gamete?
3. Plant Reproduction	Define pollination.
4. Growth and Development	Define growth.
5. Hormones and Endocrine System	What is an endocrine gland?
6. Human Reproduction	Where are sperm produced?
7. Plant Reproduction	Where is pollen produced?
8. Growth and Development	Define development.
9. Hormones and Endocrine System	How are hormones transported?
10. Human Reproduction	Where do eggs develop?
11. Plant Reproduction	What is an ovule?
12. Growth and Development	Which type of cell division increases body cell number?

Writing Task

Write a short paragraph of 80-100 words using at least four biological concepts reviewed on this page. Underline the four target scientific terms.

Self-Assessment

- Score 10-12: Strong mastery.
- Score 7-9: Good progress; review minor errors.
- Score 4-6: Revisit the related lessons.
- Score 0-3: Study the lesson explanations before retrying.

Writing Task Response

REVISION 07

Review Test 7 - Cell Division, Genetics & Evolution

Mixed practice - complete without looking back at the lessons.

1. DNA, Genes and Chromosomes	What is DNA?
2. Cell Division: Mitosis and Meiosis	What is mitosis?
3. Genetics and Inheritance	Define genotype.
4. Variation and Natural Selection	Define variation.
5. Evolution	Define evolution.
6. DNA, Genes and Chromosomes	Define a gene.
7. Cell Division: Mitosis and Meiosis	What is meiosis?
8. Genetics and Inheritance	Define phenotype.
9. Variation and Natural Selection	Give one example of continuous variation.
10. Evolution	What is a fossil?
11. DNA, Genes and Chromosomes	What is a chromosome?
12. Cell Division: Mitosis and Meiosis	How many daughter cells usually result from mitosis?

Writing Task

Write a short paragraph of 80-100 words using at least four biological concepts reviewed on this page. Underline the four target scientific terms.

Self-Assessment

- Score 10-12: Strong mastery.
- Score 7-9: Good progress; review minor errors.
- Score 4-6: Revisit the related lessons.
- Score 0-3: Study the lesson explanations before retrying.

Writing Task Response

REVISION 08

Review Test 8 - Ecology, Conservation & Integrated Biology

Mixed practice - complete without looking back at the lessons.

1. Ecology and Ecosystems	Define population.
2. Food Chains, Food Webs and Nutrient Cycles	What is a producer?
3. Human Impact, Conservation and Biology Review	Define biodiversity.
4. Ecology and Ecosystems	Define community.
5. Food Chains, Food Webs and Nutrient Cycles	What is a decomposer?
6. Human Impact, Conservation and Biology Review	Define conservation.
7. Ecology and Ecosystems	Define ecosystem.
8. Food Chains, Food Webs and Nutrient Cycles	What do arrows in a food chain show?
9. Human Impact, Conservation and Biology Review	Give three major threats to biodiversity.
10. Ecology and Ecosystems	Give two abiotic factors.
11. Food Chains, Food Webs and Nutrient Cycles	Why does energy decrease at higher trophic levels?
12. Human Impact, Conservation and Biology Review	What is sustainable use?

Writing Task

Write a short paragraph of 80-100 words using at least four biological concepts reviewed on this page. Underline the four target scientific terms.

Self-Assessment

- Score 10-12: Strong mastery.
- Score 7-9: Good progress; review minor errors.
- Score 4-6: Revisit the related lessons.
- Score 0-3: Study the lesson explanations before retrying.

Writing Task Response

ANSWER KEY

Answer Key - Part 5: Practical & Data Skills

A compact reference for scientific investigation and interpretation.

Core Practical Skills

Concept	Explanation	Example
Independent variable	The factor deliberately changed.	
Dependent variable	The factor measured in response.	
Control variable	A factor kept constant for a fair test.	
Repeat	Repeat measurements to identify anomalies and estimate reliability.	
Mean	Add valid repeat values and divide by the number of valid values.	
Range	Highest value minus lowest value; gives simple spread.	
Anomaly	A result that does not fit the general pattern.	
Conclusion	A statement answering the question using evidence from results.	

Graph Checklist

- Independent variable on x-axis; dependent variable on y-axis.
- Use sensible linear scales and include units.
- Plot points accurately; use a best-fit line or curve when appropriate.
- Do not force a line through the origin unless scientifically justified.

Evaluation Checklist

- Identify limitations in method or measurements.
- Suggest a specific improvement linked to the limitation.
- Distinguish accuracy, precision, repeatability and reproducibility.
- Do not claim causation if the evidence shows only correlation.

Practice Prompts

1. Explain why repeats improve confidence in results.
2. State why control variables are important.
3. Describe how to calculate a percentage change.
4. Explain why an anomalous result should be investigated before exclusion.

Model Answers

- Repeats reveal variation and allow a mean to be calculated.
- Control variables help ensure changes in the dependent variable are due mainly to the independent variable.
- Percentage change = $(\text{new} - \text{original}) / \text{original} \times 100$.
- Anomalies may indicate error or real biological variation; the cause should be considered.

ANSWER KEY

Answer Key - Part 6: Command Words & Key Definitions

Use precise language to match the demand of the question.

Common Command Words

Concept	Explanation	Example
State	Give a concise fact or answer without extended reasoning.	
Describe	Give features, patterns or what happens.	
Explain	Give reasons or mechanisms; link cause to effect.	
Compare	Give similarities and differences using paired statements.	
Calculate	Show working and include units.	
Suggest	Use biological knowledge to propose a plausible answer.	
Evaluate	Consider strengths, limitations and evidence before a supported judgement.	
Predict	State an expected outcome based on a pattern or mechanism.	

Key Definitions

Diffusion	Net movement of particles from higher to lower concentration.
Osmosis	Net movement of water through a partially permeable membrane from higher to lower water potential.
Respiration	Cell reactions that release usable energy from nutrients.
Homeostasis	Maintenance of a relatively stable internal environment.
Gene	Length of DNA containing information for a functional product.
Evolution	Change in heritable population characteristics across generations.
Ecosystem	A community interacting with its physical environment.
Biodiversity	Variety of life at genetic, species and ecosystem levels.

Final Revision Routine

- Review definitions from memory.
- Redraw key diagrams without notes.
- Complete mixed questions under time pressure.
- Correct errors by writing the principle, not only the answer.

REFERENCE

Biology Reference Tables

Core equations, comparisons and scientific conventions.

Essential Word Equations

Concept	Explanation	Example
Photosynthesis	carbon dioxide + water -> glucose + oxygen	Light energy and chlorophyll required.
Aerobic respiration	glucose + oxygen -> carbon dioxide + water + energy	Mainly in mitochondria.
Anaerobic respiration in muscle	glucose -> lactic acid + energy	Less energy than aerobic respiration.
Fermentation in yeast	glucose -> ethanol + carbon dioxide + energy	Occurs without oxygen.

Mitosis vs Meiosis

Feature	Mitosis	Meiosis
Number of divisions	1	2
Typical products	2 cells	4 cells
Chromosome number	Maintained	Halved
Genetic similarity	Usually similar	Genetically varied
Main role	Growth, repair, asexual reproduction	Gamete production

Blood Vessels

Concept	Explanation	Example
Artery	Away from heart; thick elastic muscular wall; relatively high pressure.	Aorta, pulmonary artery
Vein	Toward heart; thinner wall; valves; lower pressure.	Vena cava, pulmonary vein
Capillary	One-cell-thick wall; narrow lumen; exchange surface.	Networks in tissues

Useful Mathematical Relationships

- Percentage change = $(\text{new value} - \text{original value}) / \text{original value} \times 100$.
- Magnification = image size / actual size.
- Rate = change / time.
- Surface area to volume ratio = surface area / volume.

Units

Length: mm, micrometre (µm), nm. Mass: g, mg. Volume: cm³ or mL. Time: s, min, h. Temperature: degrees C. Concentration may be given in g/dm³, mol/dm³ or percentage depending on context.

REFERENCE

Biology Quick Guide

A final one-page checklist before an examination.

Cells & Transport

- Cell membrane controls exchange.
- Diffusion: high → low concentration.
- Osmosis: water across a partially permeable membrane.
- Active transport uses energy.

Energy & Nutrition

- Photosynthesis stores light energy in glucose.
- Respiration releases usable energy.
- Enzymes are specific biological catalysts.
- Balanced diets supply nutrients, fiber and water.

Human Systems

- Alveoli are adapted for gas exchange.
- Blood transports substances and heat.
- Heart drives double circulation.
- Kidneys excrete urea and regulate water/ions.

Coordination & Reproduction

- Nervous system uses rapid impulses.
- Hormones are chemical messengers in blood.
- Fertilization forms a zygote.
- Homeostasis uses negative feedback.

Genetics & Evolution

- DNA stores genetic information.
- Mitosis maintains chromosome number.
- Meiosis produces haploid varied gametes.
- Natural selection changes populations over generations.

Ecology

- Energy flows through food chains.
- Matter cycles through ecosystems.
- Biotic and abiotic factors shape populations.
- Conservation protects biodiversity and ecosystem function.

Final Exam Checklist

- Answer exactly what the command word asks.
- Use correct scientific terms and units.
- Show calculations clearly.
- Label diagrams neatly.
- Use data as evidence in explanations.
- Review every answer for biological accuracy.